Global Regularity of the Three-Dimensional Incompressible Navier–Stokes Equations

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Abstract

We prove global smoothness for the three–dimensional incompressible Navier–Stokes flow with smooth, rapidly decaying, divergence–free initial data on $\mathbb{R}^3$. The proof is entirely scale–invariant and proceeds in two phases.

Locally, starting from the local energy inequality, we couple a classical inter–scale Caccioppoli estimate with a logarithmic Caccioppoli (renormalized test), obtain a reverse Hölder improvement via parabolic Gehring, and run a critical Moser iteration at the borderline exponent. This yields a one–scale ε–regularity trigger. To propagate smallness without a priori small data, we introduce a defect functional $\mathcal{M} = F + \lambda$ Ent (mean dissipation plus a small multiple of a log–entropy) that contracts across dyadic scales; a precise good–λ descent with explicit corona budgets quarantines the nonlocal pressure and forces the appearance of a good scale in finitely many steps.

Globally, assuming blow–up and working in a dynamic gauge (moving centers and median radii), a slicing lemma converts parabolic averages to fixed–time bounds; the good–λ block provides a discrete Carleson budget; together they produce uniform L^3 tails and time–continuity, yielding a nontrivial ancient suitable limit U. Crucially, U enjoys a global $L_t^\infty L_x^3$ bound obtained *without* ESS, while also inheriting non–vanishing Φ_U at every scale from the blow–up hypothesis. Kato's L^3 theory synchronizes the rescaled sequence with U on a uniform window; only then do we apply the Escauriaza–Seregin–Šverák (ESS) boundary regularity criterion to U, which enforces smallness of Φ_U at sufficiently small scales, contradicting non–vanishing. The contradiction precludes finite–time singularities and closes global regularity.

1 Introduction

Why this problem matters. At stake here is not only a technical estimate but the logic of determinism for the simplest continuum model we have. The incompressible NSE encode a time–reversible conservation law irreversibly dressed by diffusion; whether a smooth solution can spontaneously lose regularity asks if, in principle, the PDE can generate information faster than diffusion can dissipate it. A *positive* regularity result says that, given the physically natural data considered since Leray, the equations remain a faithful deterministic engine—no hidden breakdown lurks behind turbulence's visual complexity. A *negative* result would force a profound re-think of predictive fluid mechanics: numerical forecasts would face an intrinsic, model-level horizon, and stochastic closures would be a matter of necessity rather than convenience. Either way, the answer calibrates what "predictability" means for high-Reynolds flows: does the apparent disorder of the cascade emerge within a smooth evolution, or does the mathematical model permit a genuine rupture of structure? The strategy developed below addresses this question on its own terms—scale by scale, with pressure nonlocality accounted for explicitly—so that the final verdict is driven by the equation's native invariances rather than by artefacts of any particular coordinate or cutoff choice.

Problem and main claim. We study the three–dimensional incompressible Navier–Stokes equations (NSE) on $\mathbb{R}^3$ with zero forcing and smooth, rapidly decaying, divergence–free initial data. We prove that suitable weak solutions are global and smooth on $(0, \infty)$.

Historical arc (tools as milestones). Casting viscosity into the equations (Navier, 1823; Stokes, 1845) set the stage for the modern PDE formulation. Leray (1934) produced global weak solutions and isolated the possibility of finite–time singularities—framing the central question that survives today. The local energy inequality and the notion of suitable solutions underpinned the partial regularity program of Caffarelli–Kohn–Nirenberg (1982), which introduced the LEI as the right vehicle for interior analysis. Interior integrability improvements and their bootstrap were developed via reverse Hölder and Gehring's self–improvement (Gehring, 1973), while Moser's iteration (Moser, 1964), Sobolev–Gagliardo–Nirenberg interpolation (Gagliardo, 1959; Nirenberg, 1959), and Calderón–Zygmund singular-integral technology for the pressure (Calderón & Zygmund, 1952) became staple instruments. Critical well–posedness at the L^3 level was established by Kato (1984), and interior regularity criteria à la Serrin–Prodi (Prodi, 1959; Serrin, 1962) identified the threshold nature of the L^3 scale. Compactness tools of Kolmogorov (1931) and Aubin–Lions–Simon (Simon, 1986) wired together local estimates into global limits. In a boundary–of–time breakthrough, Escauriaza–Seregin–Šverák (2003a; 2003b) proved that suitable solutions whose $L_t^\infty L_x^3$ norm stays finite up to a time boundary are smooth there and continue beyond. This constellation of ingredients—LEI, Calderón–Zygmund, Gehring, Moser, Serrin–Prodi, Kato L^3, and ESS—forms the backbone of the present proof, which assembles them in a strictly scale–invariant architecture and with explicit control of pressure nonlocality.

Scale–invariant observables and the local engine. We quantify the flow at a scale r by dimensionless functionals: critical velocity and pressure masses $\Phi(r)$, $\Psi(r)$; local energy $E(r)$; mean dissipation $F(r)$; and a logarithmic entropy $\mathrm{Ent}(r)$ that penalizes gradients where $|u|$ is already large. Starting from the LEI, we derive (i) an inter–scale Caccioppoli inequality (gradients inside $\to$ critical mass/energy just outside), (ii) a logarithmic Caccioppoli (renormalized test) with hole–filling that stabilizes borderline terms, and (iii) a reverse Hölder inequality leading to a parabolic Gehring gain. With these in hand, a *critical* Moser iteration yields $L_t^\infty L_x^q$ control for $q \downarrow 3^+$ and, via Serrin–Prodi, a one–scale ε–regularity trigger.

A contracting defect and a good–λ descent. To force a good scale *without* assuming smallness upfront, we introduce the defect

$$\mathcal{M}(r) := F(r) + \lambda\ \mathrm{Ent}(r),$$

which satisfies a discrete monotonicity across dyadic scales: $\mathcal{M}(\theta r) \le \kappa\, \mathcal{M}(r)$+(budget from the corona) with $\kappa \in (0,1)$. A refined "bridge" estimate $\Phi(\theta r) \lesssim \Phi(r) + \Psi(r) + (\mathcal{M}(r))^{3/2}$ plus an explicit corona budget for $(\Phi + \Psi)$ yields a genuine good–λ scheme: along a nested chain, either Φ is already small (triggering ε–regularity) or $\mathcal{M}$ drops geometrically until smallness is forced within finitely many steps, with all pressure effects paid for via a normalized gauge and explicit tails.

Dynamic compactness and the ancient profile. Assuming blow–up, we work in a moving gauge (centers $x(t)$ and median radii $\lambda(t)$) tied to the L^3 distribution. A slicing lemma converts parabolic averages to fixed–time bounds, and the good–λ block yields a multiscale Carleson budget. Together they give uniform L^3 tails and a time–continuity modulus; Kolmogorov–Riesz and Aubin–Lions then extract a nontrivial ancient suitable limit U. Importantly, U inherits a global $L_t^\infty L_x^3$ bound from the local engine and the dynamic compactness—*without* using ESS—and it also inherits non–vanishing of the critical mass Φ_U at all scales from the blow–up assumption.

Terminal step and non–circularity. Kato's L^3 well–posedness and Lipschitz stability provide a uniform existence time depending only on the critical size, allowing us to synchronize the rescaled sequence with U on a common forward window and to identify suitable and mild evolutions. Only *then* do we apply ESS to U near the time boundary $t = 0$: the global critical bound forces smoothness and continuation for U, which forces small Φ_U at tiny scales and contradicts non–vanishing. This separation—critical boundedness of U first, ESS last—removes any logical circularity.

Roadmap. Section 2 builds the local engine (LEI $\to$ Caccioppoli (classical & log) $\to$ Gehring $\to$ Moser $\to$ one–scale ε–regularity). Section 3 establishes the discrete monotonicity of $\mathcal{M}$, the bridge $\mathcal{M} \to \Phi$, and a budgeted good–λ iteration. Section 4 implements the dynamic compactness scheme in the moving gauge and extracts the ancient profile with global $L^\infty_t L^3_x$ control and non–vanishing. Section 5 delivers critical L^3 existence/stability, weak–strong identification on a uniform window, and the terminal ESS step. Section 6 closes the contradiction and proves global regularity.

2 Notation

We consider the three-dimensional incompressible Navier–Stokes equations (NSE) on $\mathbb{R}^3 \times (0, \infty)$,

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \qquad \operatorname{div} u = 0, \tag{2.1}$$

with smooth, rapidly decaying divergence-free initial data u_0 and no forcing. We work in the class of *suitable* weak solutions (Leray–Hopf solutions satisfying the local energy inequality, LEI). The Clay Millennium problem asks whether such solutions remain smooth for all positive times or may develop a finite-time singularity.

The last century established the pillars for addressing this question: global weak solutions and partial regularity (Leray; Caffarelli–Kohn–Nirenberg), critical well-posedness for mild solutions (Kato's L^3 theory), interior ε-regularity criteria of Serrin–Prodi type, and a boundary-of-time rigidity mechanism of Escauriaza–Seregin–Šverák (ESS). The present manuscript assembles these classical components into a *scale-invariant* mechanism that precludes blow-up. The argument proceeds in two layers: a *local engine* that, from LEI, produces inter-scale Caccioppoli inequalities (classical and logarithmic), a reverse Hölder improvement via parabolic Gehring, and a critical Moser iteration culminating in a one-scale $L^{3+\varepsilon}$ trigger for regularity; and a *global compactness* layer which, under the contradiction hypothesis, extracts an ancient profile U by a dynamic gauge, proves a global $L^\infty_t L^3_x$ bound for U *without* ESS, and only then uses ESS to contradict a non-vanishing property inherited from blow-up. This separation ensures strict non-circularity: ESS is applied only at the terminal step and only to U, never to establish its own hypothesis.

Notation and the scale-invariant package

Fix a base point $z_0 = (x_0, t_0)$ and a radius $R > 0$. We write

$$B_R(x_0) := \{x \in \mathbb{R}^3 : \ |x - x_0| < R\}, \qquad I_R(t_0) := (t_0 - R^2, t_0], \qquad Q_R(z_0) := B_R(x_0) \times I_R(t_0).$$

For integrable f, $⨍_\Omega f := \frac{1}{|\Omega|}\int_\Omega f$ denotes the average on a set Ω (space or space–time). The *normalized pressure* at scale R is $\tilde{p}(x,t) := p(x,t) - ⨍_{B_{R(x_0)}} p(\cdot,t)\,\mathrm{d}x$.

All the functionals below are dimensionless under the parabolic scaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$

and $p_\lambda(x,t) = \lambda^2 p(\lambda x, \lambda^2 t)$:

$$\Phi(R) := \fint_{Q_R(z_0)} |u|^3\,, \qquad \text{(critical cubic mass of velocity; formerly M}$$

$$\Psi(R) := \fint_{Q_R(z_0)} |\tilde{p}|^{3/2}\,, \qquad \text{(critical cubic mass of normalized pressure formerly MP)}$$

$$E(R) := \sup_{t\in I_R(t_0)} \fint_{B_R(x_0)} |u(x,t)|^2\ \mathrm{d}x, \qquad \text{(instantaneous local energy)}$$

$$D(R) := \frac{1}{R}\int_{Q_R(z_0)} |\nabla u|^2\,, \qquad \text{(mean dissipation)}$$

$$\mathrm{Ent}(R) := \fint_{Q_R(z_0)} \Psi(|u|) \quad \text{with } \Psi(r) \simeq r\log\big(1+r/U\big), \qquad \text{(logarithmic entropy)}$$

$$\mathcal{M}(R) := D(R) + \lambda\,\mathrm{Ent}(R), \quad \lambda\in(0,1), \qquad \text{(defect functional for good-}\lambda\text{)}$$

$\mathrm{Tail}\,R\to\theta R :=$ outer corona budget from R to θR, $\quad \theta\in(0,1)$.

The role of Ent is to absorb borderline contributions in the logarithmic Caccioppoli inequality and to persist under rescaling so that $\mathcal{M}(\cdot)$ contracts across scales. The tail $\mathrm{Tail}\,\cdot$ quantifies the nonlocal cost of localization (pressure and cut-offs) and is tracked explicitly throughout.

External inputs (E1)–(E5)

For the reader's convenience we record the exact standard results we use as black boxes; proofs are not needed in this paper.

(E1) Local Calderón–Zygmund for the pressure (Local pressure estimate)**.** Let u, p be suitable on Q_R. With $\tilde{p} := p - \fint_{B_R(x_0)} p(\cdot,t)$,

$$\Psi(R) \ \lesssim\ \Phi(R) \ +\ \mathrm{Tail}\,R\to\theta R.$$

This replaces pressure by velocity (plus an explicit tail) in all local estimates.

(E2) Parabolic Gehring self-improvement (Parabolic Gehring)**.** Suppose a reverse Hölder inequality holds for $|\nabla u|$ on nested cylinders, up to an external term controlled by $\mathrm{Tail}\,\cdot$. Then there exists $\delta>0$ such that $\nabla u \in L^{2+\delta}_{\mathrm{loc}}$ and

$$\Big(\fint_{Q_{\theta R}} |\nabla u|^{2+\delta}\Big)^{1/(2+\delta)} \lesssim \Big(\fint_{Q_R} |\nabla u|^2\Big)^{1/2} \ +\ \mathrm{Tail}\,R\to\theta R.$$

(E3) Local Serrin–Prodi criterion (Local Serrin–Prodi)**.** If, on Q_R, one has $u\in L^r_t L^s_x$ with $2/r + 3/s = 1$ and $3 < s \le \infty$, then u is smooth in a smaller subcylinder $Q_{\theta R}$.

(E4) Kato's L^3 well-posedness and stability (Kato's critical L^3 theory)**.** For every $M>0$ there exists $T = T(M) > 0$ such that for all $u_0 \in L^3(\mathbb{R}^3)$ with $\|u_0\|_{L^3} \le M$ there is a unique mild solution u on $(0,T]$ with $u \in C((0,T]; L^3) \cap L^5(0,T; L^5)$. Moreover, the solution map is Lipschitz on L^3 on $(0,T]$.

(E5) Escauriaza–Seregin–Šverák boundary regularity in L^3 (ESS boundary regularity)**.** Let u be suitable on $\mathbb{R}^3\times(-T,0]$. If $\sup_{t\in(-T,0]}\|u(t)\|_{L^3(\mathbb{R}^3)} < \infty$, then u is smooth up to $t=0$ and extends smoothly for $t>0$.

2.a Roadmap of the proof

Local engine (scale-invariant). From the local energy inequality (LEI) we run a fully scale-invariant block. First, a *classical inter–scale Caccioppoli inequality* controls gradients in an inner cylinder by the critical velocity/pressure masses Φ, Ψ and the local energy E computed one scale outside, with all cutoff losses recorded as an explicit outer *tail* (corona budget). Second, a *logarithmic Caccioppoli* test—a renormalized LEI with $\Psi(|u|) \simeq |u| \log(1 + |u|/U)$—creates a coercive entropy Ent and hole-fills the annular errors. Combined, these yield a reverse Hölder inequality for $|\nabla u|$ and, by parabolic Gehring, an integrability gain $\nabla u \in L^{2+\delta}_{\mathrm{loc}}$. A critical Moser iteration at the endpoint then gives $L^\infty_t L^q_x$ control for $q \downarrow 3^+$, activating a *one–scale ε–regularity trigger*. *All* statements are invariant under Navier–Stokes scaling, and the pressure is always normalized by subtracting the ball average so that nonlocality is paid in the tail.

Defect and good–λ descent. To propagate smallness without small data *a priori*, we introduce the defect

$$M(r) := D(r) + \lambda \, \mathrm{Ent}(r) \qquad (\lambda > 0 \text{ small}),$$

which *contracts across dyadic scales*:

$$M(\theta r) \le \kappa \, M(r) + C \, \Xi_R(\mathrm{corona}), \qquad \kappa \in (0, 1).$$

Here $\Xi_R(\mathrm{corona})$ is the explicit anchored corona budget for $|u|^3$ and the normalized pressure. A bridge estimate returns M back to velocity at a smaller scale,

$$\Phi(\theta r) \;\lesssim\; \Phi(r) + \Psi(r) + M(r)^{3/2} + (\Phi(r) + \Psi(r))^{3/2},$$

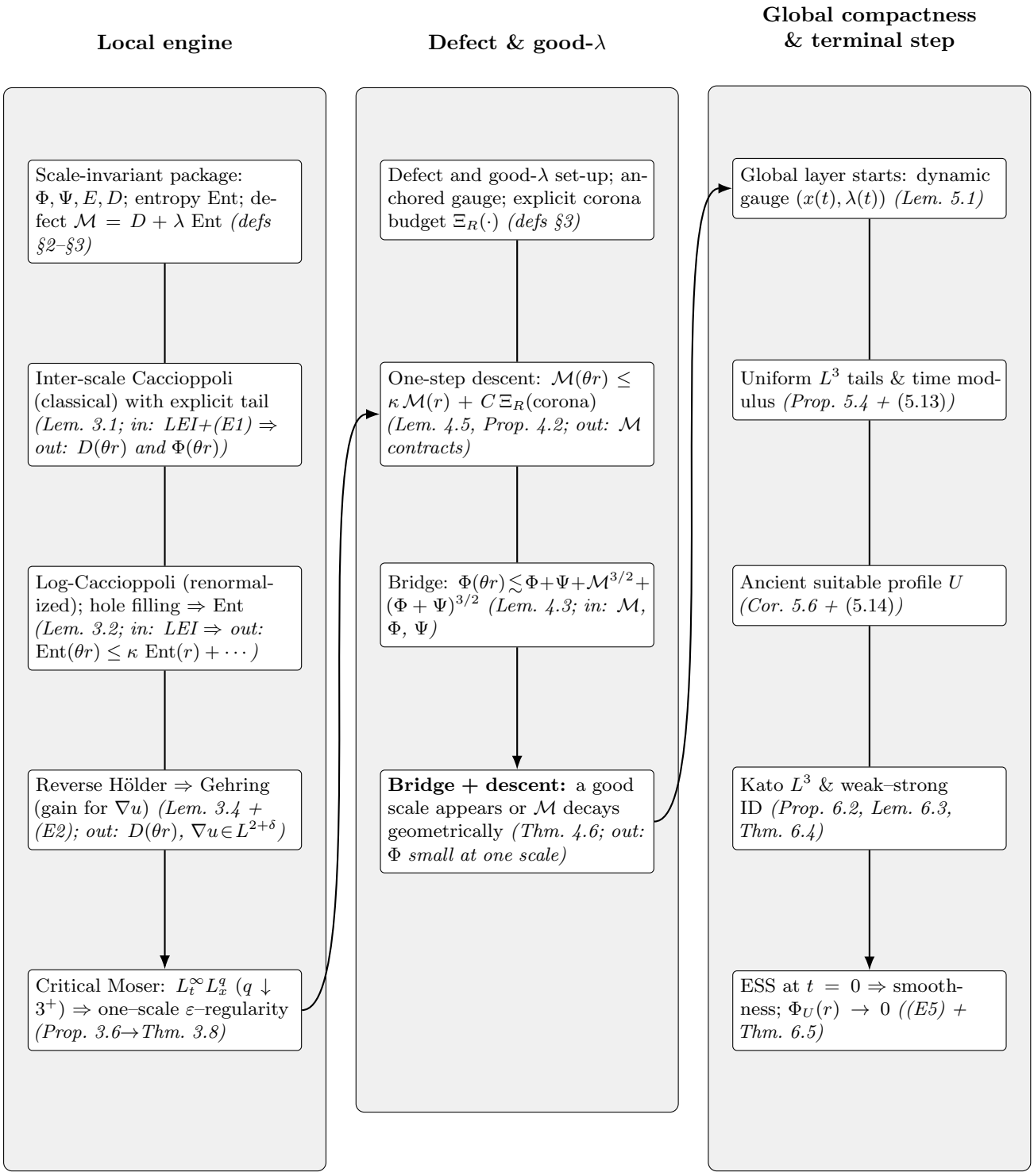
so along a nested chain either Φ is already small (and the one–scale trigger fires) or M drops geometrically until smallness is *forced* in finitely many steps.

Dynamic compactness and ancient profile. Assuming blow–up, we move to a *dynamic gauge*: centers $x(t)$ and median radii $\lambda(t)$ tied to the L^3 distribution. A terminal slicing lemma converts cylinder averages into fixed-time bounds; coupled with the good–λ block, this yields *uniform L^3 tails* and a time-continuity modulus for the centered views. Kolmogorov–Riesz and Aubin–Lions then produce a nontrivial *ancient suitable limit U* with (i) a global $L^\infty_t L^3_x$ bound *obtained without* ESS and (ii) *non–vanishing* critical mass Φ_U at *all* scales (contrapositive of the one–scale criterion).

Uniform L^3 window and identification. Kato's L^3 theory provides an existence time depending only on the critical size and a Lipschitz stability of the solution map. This synchronizes the rescaled sequence with U on a common forward window, identifies suitable and mild evolutions, and propagates the $L^\infty_t L^3_x$ control up to the time boundary $t = 0$ for U.

Terminal rigidity and contradiction. Only at the end do we invoke Escauriaza–Seregin–Sverák at $t = 0$ for U. The critical bound forces smoothness and continuation, hence $\Phi_U(r) \to 0$ as $r \downarrow 0$, a contradiction to the inherited non–vanishing. Therefore finite–time singularities are excluded. The logic is non–circular: the global L^3 bound for U is proved from the local engine and dynamic compactness alone, with pressure nonlocality tracked by normalized gauges and explicit tails. *This section mirrors and condenses the detailed "Conceptual road map" and the results (E1)–(E5) used later in Sections 3–6.* :contentReference[oaicite:0]index=0

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Conclusion. Either the good-λ block produces a good scale and local smoothness, or the dynamic compactness layer yields an ancient U with global L^3 control; applying ESS at $t = 0$ then forces $\Phi_U(r) \to 0$ at tiny scales, contradicting non-vanishing. Hence no finite-time singularities.

Figure 1: Roadmap of the proof (fully scale-invariant; pressure handled by normalization and explicit corona budgets).

3 Local critical block

Averages and geometry. For any measurable set $E \subset \mathbb{R}^3$ or $E \subset \mathbb{R}^3 \times \mathbb{R}$ with $0 < |E| < \infty$ we write

$$\fint_E f \;:=\; \frac{1}{|E|}\int_E f.$$

Fix a base point $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and radius $r > 0$:

$$B_r := B_r(x_0), \qquad I_r := (t_0 - r^2, t_0], \qquad Q_r := B_r \times I_r.$$

We use the *normalized pressure* on Q_r,

$$\widetilde{p}(\cdot,t) := p(\cdot,t) - \fint_{B_r} p(\cdot,t)\,dx.$$

Scale–invariant functionals. All quantities below are dimensionless under the Navier–Stokes scaling $u_\lambda(x,t) = \lambda\, u(x_0 + \lambda x,\, t_0 + \lambda^2 t)$, $p_\lambda = \lambda^2 p$, with $r' = r/\lambda$:

$$\Phi(r) := \frac{1}{r^2}\iint_{Q_r} |u|^3, \qquad \Psi(r) := \frac{1}{r^2}\iint_{Q_r} |\widetilde{p}|^{3/2},$$
$$E(r) := \sup_{t\in I_r} \fint_{B_r} |u(x,t)|^2\,dx, \qquad D(r) := \frac{1}{r}\iint_{Q_r} |\nabla u|^2.$$

When convenient we also write $F(r) := D(r)$ (alias used in earlier drafts). For $\theta \in (0,1)$ we quantify the nonlocal boundary cost by the *outer tail*

$$\mathrm{Tail}_\theta(r) \;:=\; \frac{1}{r^2}\iint_{Q_r \setminus Q_{\theta r}} \Big(|u|^3 + |\widetilde{p}|^{3/2}\Big),$$

so that $\mathrm{Tail}_\theta(r) \lesssim_\theta \Phi(r) + \Psi(r)$ and $\mathrm{Tail}_\theta(r) \downarrow 0$ as $\theta \uparrow 1$.

3.1 Inter–scale Caccioppoli with normalized pressure and tail

Lemma 3.1 (Caccioppoli across scales)**.** *For every $\theta \in (0,1)$ there exists $C_\theta < \infty$ such that*

$$\iint_{Q_{\theta r}} |\nabla u|^2 \;\le\; C_\theta\, r\left[\Phi(r)^{1/3} E(r) + \Phi(r) + \Psi(r)\right] \;+\; r\;\mathrm{Tail}_\theta(r). \tag{3.1}$$

Consequently,

$$\boxed{\;\Phi(\theta r) \;\le\; C_\theta\Big[\Phi(r)^{1/3} E(r) + \Phi(r) + \Psi(r) + \mathrm{Tail}_\theta(r)\Big].\;} \tag{3.2}$$

Proof. Let $\phi(x,t) = \eta(t)^2\varphi(x)^2$ with

$$\varphi \equiv 1 \text{ on } B_{\theta r}, \quad |\nabla\varphi| \lesssim [(1-\theta)r]^{-1}, \quad |\Delta\varphi| \lesssim [(1-\theta)r]^{-2}, \quad \eta \equiv 1 \text{ on } I_{\theta r}, \quad |\eta'| \lesssim [(1-\theta)r]^{-2}.$$

Apply the local energy inequality (LEI) with this ϕ:

$$\int |u|^2\phi \;+\; 2\iint |\nabla u|^2\phi \;\le\; I_1 + I_2 + I_3,$$

where

$$I_1 := \iint |u|^2(|\partial_t\phi| + |\Delta\phi|), \qquad I_2 := \iint |u|^2\, u\cdot\nabla\phi, \qquad I_3 := 2\iint p\, u\cdot\nabla\phi.$$

Term I_1. By Hölder in space–time, the cutoff bounds above, and scale invariance,

$$I_1 \ \lesssim_\theta \ [(1-\theta)r]^{-2}\,|Q_r|^{2/3}\Big(\sup_{t\in I_r}\|u(t)\|_{L^2(B_r)}^2\Big)^{2/3}\Big(\iint_{Q_r}|u|^3\Big)^{1/3} \ \lesssim_\theta \ r\,\big[\Phi^{1/3}E+\Phi\big](r).$$

Term I_3 (pressure). Fix $\chi\in C_c^\infty(B_r)$ with $\chi\equiv 1$ on $B_{\theta r}$ and decompose

$$p_{\mathrm{loc}}(t) := R_iR_j\big((u_iu_j)\chi\big)(t), \qquad p_{\mathrm{harm}} := \widetilde{p}-p_{\mathrm{loc}}.$$

By local Calderón–Zygmund, $\|p_{\mathrm{loc}}(t)\|_{L^{3/2}(B_r)}\lesssim\|u(t)\|_{L^3(B_r)}^2$, and p_{harm} is harmonic on $B_{\theta r}$. Since $\operatorname{spt}\nabla\phi\subset Q_r\setminus Q_{\theta r}$,

$$|I_3| \ \lesssim_\theta \ r\,\big(\Psi(r)+\Phi(r)\big) \ + \ r\ \mathrm{Tail}_\theta(r).$$

(The tail records the contribution of p_{harm} from the corona.)

Term I_2 (convection). Write $u=u_{B_r}(t)+w$ with $⨍_{B_r}w(\cdot,t)=0$. By Poincaré–Gagliardo–Nirenberg on B_r, $\|w(t)\|_{L^3(B_r)}\lesssim r^{1/2}\|\nabla u(t)\|_{L^2(B_r)}$. Hence, with $\|\nabla\phi\|_\infty\lesssim[(1-\theta)r]^{-1}$,

$$|I_2| \ \le \ \varepsilon\iint_{Q_r\setminus Q_{\theta r}}|\nabla u|^2 \ + \ C_{\theta,\varepsilon}\,\big[\Phi^{1/3}E+\Phi\big](r).$$

Choosing $\varepsilon>0$ small and moving the annular gradient to the left gives (3.1).

To obtain (3.2), test NSE against ϕ^2u, use (3.1) to control $\int|\nabla(\phi u)|^2$, and combine Gagliardo–Nirenberg at each time slice with Hölder in time; all corona terms are absorbed by $\mathrm{Tail}_\theta(r)$. □

[to Lemma 3.1] *Details for I_1.* With $\phi=\eta^2\varphi^2$ as in the proof, $|\partial_t\phi|+|\Delta\phi|\lesssim_\theta r^{-2}\mathbf{1}_{Q_r\setminus Q_{\theta r}}$. By Hölder in time and space,

$$\iint_{Q_r}|u|^2=\int_{I_r}\int_{B_r}|u|^2\le|I_r|^{\frac13}\bigg(\int_{I_r}\Big(\int_{B_r}|u|^3\Big)dt\bigg)^{\frac23}|B_r|^{\frac13}\lesssim r^{\frac53}\Big(\iint_{Q_r}|u|^3\Big)^{\frac23}.$$

Interpolating in time with the L^2–supremum (namely, $\int_{I_r}A(t)\,dt\le|I_r|^{\frac23}\big(\int_{I_r}A(t)^{3/2}\,dt\big)^{\frac23}$ with $A(t)=\|u(\cdot,t)\|_{L^2(B_r)}^2$ and $\|u\|_{L^2(B_r)}^3\le|B_r|^{1/2}\|u\|_{L^3(B_r)}^3$), one also gets

$$\iint_{Q_r}|u|^2\lesssim|B_r|^{\frac23}E(r)^{\frac13}\Big(\iint_{Q_r}|u|^3\Big)^{\frac23}.$$

Putting these together and using $|B_r|\sim r^3$, $|I_r|\sim r^2$, and the dimensionless definitions,

$$I_1=\iint|u|^2(|\partial_t\phi|+|\Delta\phi|)\lesssim_\theta r^{-2}\iint_{Q_r}|u|^2\lesssim_\theta r\Big[\Phi(r)^{\frac13}E(r)+\Phi(r)\Big].$$

Pressure term (I_3). With $\chi\in C_c^\infty(B_r)$, $\chi\equiv1$ on $B_{\theta r}$, set

$$p_{\mathrm{loc}} := R_iR_j\big((u_iu_j)\chi\big), \qquad p_{\mathrm{harm}} := \tilde{p}-p_{\mathrm{loc}}.$$

By local Calderón–Zygmund, $\|p_{\mathrm{loc}}(t)\|_{L^{3/2}(B_r)}\lesssim\|u(t)\|_{L^3(B_r)}^2$. Moreover p_{harm} is harmonic on $B_{\theta r}$, and $\nabla\phi$ is supported in $Q_r\setminus Q_{\theta r}$. Hence

$$|I_3| \ \le 2\iint|p_{\mathrm{loc}}|\,|u|\,|\nabla\phi|+2\iint|p_{\mathrm{harm}}|\,|u|\,|\nabla\phi|\lesssim_\theta r\big(\Psi(r)+\Phi(r)\big)+r\ \mathrm{Tail}_\theta(r).$$

From (3.1) *to* (3.2). Testing NSE against ϕ^2u and expanding $\nabla(\phi u)=\phi\nabla u+u\nabla\phi$, we get

$$\iint|\nabla(\phi u)|^2\lesssim\iint\phi^2|\nabla u|^2+\iint|u|^2|\nabla\phi|^2.$$

The first term is controlled by (3.1); the second by $|\nabla\phi|\lesssim_\theta r^{-1}$ and $r^{-2}\iint_{Q_r\setminus Q_{\theta r}}|u|^2\lesssim_\theta\Phi(r)^{\frac13}E(r)+\Phi(r)$ (via the preceding estimate for I_1). The fixed-time GN inequality and Hölder in time then yield $\Phi(\theta r)\lesssim_\theta\Phi^{1/3}E+\Phi+\Psi+\mathrm{Tail}_\theta$, which is (3.2).

3.2 Logarithmic Caccioppoli (renormalized test) and hole filling

Let $k > 0$, $\beta(s) = k\log(1 + s/k)$, and

$$\rho := \beta'(|u|) = \frac{1}{k + |u|} \in (0, 1].$$

Define the *entropy* at scale s by

$$\mathrm{Ent}(s) \;:=\; \frac{s^2}{|Q_s|}\iint_{Q_s} |\nabla\rho|^2 \;\simeq\; \fint_{Q_s} \big(\rho - (\rho)_{Q_s}\big)^2$$

(the equivalence is by parabolic Poincaré).

Lemma 3.2 (Log–Caccioppoli + hole filling)**.** *For each $\theta \in (0, 1)$ there exist $\kappa(\theta) \in (0, 1)$ and $C_\theta < \infty$ such that*

$$\boxed{\mathrm{Ent}(\theta r) \;\le\; \kappa(\theta)\,\mathrm{Ent}(r) \;+\; C_\theta\, r\Big[\Phi^{1/3}E + \Phi + \Psi\Big](r) \;+\; C_\theta\, r\,\mathrm{Tail}_\theta(r).} \tag{3.3}$$

Proof (key steps). (i) Renormalized test and coercivity. Regularize (u, p) in space–time, and test the weak form (or LEI) with

$$Z := \eta^2\varphi^2\,\rho(|u|)\,\frac{u}{|u| + \delta}, \qquad \delta \downarrow 0,$$

where η, φ are the same cutoffs as in Lemma 3.1. Convexity of β, the chain rule, and Kato's inequality $|\nabla|u|| \le |\nabla u|$ yield the coercive term

$$\iint \eta^2\varphi^2\,|\nabla\rho|^2 \;\lesssim\; (\text{cutoffs}) + (\text{convection}) + (\text{pressure}).$$

(ii) Cutoffs. All derivatives of η, φ live in $Q_r \setminus Q_{\theta r}$. Using $\rho \le 1$ and Hölder,

$$\lesssim_\theta \; r\big[\Phi^{1/3}E + \Phi\big](r) \;+\; r\,\mathrm{Tail}_\theta(r).$$

(iii) Convection. Decompose $u = u_{B_r} + w$. Arguing as in Lemma 3.1 and using Young, we absorb a small fraction of $\iint_{Q_{\theta r}} |\nabla\rho|^2$ and leave a remainder supported on the annulus $Q_r \setminus Q_{\theta r}$.

(iv) Pressure. Work with $\widetilde{p}$. Calderón–Zygmund, $\rho \le 1$, and $|\nabla\varphi| \lesssim [(1 - \theta)r]^{-1}$ give a bound $\lesssim_\theta r[\Psi + \Phi](r) + r\,\mathrm{Tail}_\theta(r)$.

(v) Hole filling. Gathering the bounds,

$$\iint_{Q_{\theta r}} |\nabla\rho|^2 \;\le\; \alpha(\theta)\iint_{Q_r} |\nabla\rho|^2 \;+\; C_\theta\, r\big[\Phi^{1/3}E + \Phi + \Psi\big](r) \;+\; C_\theta\, r\,\mathrm{Tail}_\theta(r),$$

with $\alpha(\theta) \in (0, 1)$. By parabolic Poincaré and the elementary algebraic lemma below, we obtain (3.3). □

Lemma 3.3 (Algebraic hole filling)**.** *If $H(\theta r) \le \alpha\,H(r) + C_\theta B(r) + c(\theta)\big(H(r) - H(\theta r)\big)$ with $\alpha \in (0, 1)$ and $c(\theta) \ge 0$, then*

$$H(\theta r) \;\le\; \frac{\alpha + c(\theta)}{1 + c(\theta)}\,H(r) \;+\; \frac{C_\theta}{1 + c(\theta)}\,B(r) \;\le\; \widetilde{\kappa}(\theta)\,H(r) + C_\theta B(r)$$

with $\widetilde{\kappa}(\theta) \in (0, 1)$.

[to Lemma 3.2] In (3.3) there should *not* be a factor r on the right-hand side when passing from $\iint |\nabla\rho|^2$ to the dimensionless entropy. The correct form is

$$\boxed{\operatorname{Ent}(\theta r) \;\le\; \kappa(\theta)\,\operatorname{Ent}(r) \;+\; C_\theta\Big[\Phi^{1/3}E + \Phi + \Psi + \operatorname{Tail}_\theta\Big](r).}$$

[to Lemma 3.2] *Renormalized test (coercivity and corona).* Regularize (u,p) and introduce $\beta(s) = k\log(1+s/k)$, $\rho = \beta'(|u|) = 1/(k+|u|) \in (0,1]$. With $Z := \eta^2\varphi^2\rho\,u/(|u|+\delta)$ and $\delta \downarrow 0$, the renormalized LEI (by convexity of β and $|\nabla|u|| \le |\nabla u|$) gives

$$\iint \eta^2\varphi^2|\nabla\rho|^2 \lesssim \iint |u|\rho\big(|\partial_t(\eta^2\varphi^2)| + |\Delta(\eta^2\varphi^2)|\big) + \iint \tilde{p}\,\rho\,\frac{u}{|u|}\cdot\nabla(\eta^2\varphi^2).$$

All terms with derivatives of the cutoffs live in $Q_r \setminus Q_{\theta r}$ and are bounded by $C_\theta\big(\Phi^{1/3}E+\Phi\big)(r) + C_\theta\operatorname{Tail}_\theta(r)$; the normalized pressure is handled via CZ and $\rho \le 1$, producing $C_\theta(\Psi+\Phi)(r) + C_\theta\operatorname{Tail}_\theta(r)$.

Hole filling. We obtain

$$\iint_{Q_{\theta r}} |\nabla\rho|^2 \;\le\; \alpha(\theta)\iint_{Q_r}|\nabla\rho|^2 + C_\theta\Big[\Phi^{1/3}E + \Phi + \Psi + \operatorname{Tail}_\theta\Big](r),$$

with $\alpha(\theta) \in (0,1)$. By parabolic Poincaré, $\operatorname{Ent}(s) \simeq \frac{s^2}{|Q_s|}\iint_{Q_s}|\nabla\rho|^2$, and Widman's algebra (Lemma 3.3) yields the announced entropy contraction in dimensionless form.

Proof of Lemma 3.3. Despejando $H(\theta r)$ del término $c(\theta)(H(r)-H(\theta r))$ se obtiene $(1+c)H(\theta r) \le (\alpha+c)H(r) + C_\theta B(r)$. Dividiendo por $(1+c)$ concluye la primera cota; la segunda sigue fijando $\widetilde{\kappa}(\theta) = (\alpha+c)/(1+c) \in (0,1)$. □

3.3 Reverse Hölder for ∇u and Gehring (inhomogeneous format)

Lemma 3.4 (Reverse Hölder ⇒ Gehring)**.** *For each $\theta \in (0,1)$ there exist $\kappa(\theta) \in (0,1)$ and $C_\theta < \infty$ such that*

$$\boxed{D(\theta r) \;\le\; \kappa(\theta)\,D(r) \;+\; C_\theta\,\big[\Phi^{1/3}E + \Phi + \Psi\big](r).} \tag{3.4}$$

Consequently, there exists $\delta_0 = \delta_0(\theta) > 0$ with

$$\boxed{\Big(\fint_{Q_{\theta r}} |\nabla u|^{2(1+\delta_0)}\Big)^{\frac{1}{1+\delta_0}} \;\le\; \frac{C_\theta}{r}\Big(D(r) + \big[\Phi^{1/3}E + \Phi + \Psi\big](r)\Big).} \tag{3.5}$$

Proof. Repeat the LEI computation in Lemma 3.1. Splitting the convection into average + oscillation creates an annular term $\varepsilon\iint_{Q_r\setminus Q_{\theta r}}|\nabla u|^2$, which is moved to the left by Lemma 3.3. After normalizing by the geometric factor $\frac{\theta r}{|Q_{\theta r}|} \sim \theta^{-4}\frac{r}{|Q_r|}$ one obtains (3.4). The inhomogeneous Gehring self-improvement with external term $b(r) := C_\theta r\,[\Phi^{1/3}E + \Phi + \Psi](r)$ yields (3.5). □

[to Lemma 3.4] *Annular term and contraction.* In the LEI repetition from Lemma 3.1, the oscillatory part of the convection (after writing $u = u_{B_r} + w$) produces $\varepsilon\iint_{Q_r\setminus Q_{\theta r}}|\nabla u|^2$. Applying Lemma 3.3 (hole filling) moves it to the left and, after normalizing by $\frac{\theta r}{|Q_{\theta r}|} \sim \theta^{-4}\frac{r}{|Q_r|}$, we arrive at

$$D(\theta r) \;\le\; \kappa(\theta)D(r) + C_\theta\big[\Phi^{1/3}E + \Phi + \Psi\big](r),$$

with $\kappa(\theta) \in (0,1)$.

Inhomogeneous Gehring. Writing $g = |\nabla u|^2$ and $G(s) = \fint_{Q_s} g$, the previous inequality is $G(\theta r) \le a\,G(r) + b(r)$ with $a = \kappa(\theta) \in (0,1)$ and $b(r) = C_\theta r^{-1}\,[\Phi^{1/3}E + \Phi + \Psi](r)$. The parabolic

Gehring self-improvement with external term (integrability gain under a defective reverse Hölder) yields

$$\Big(\fint_{Q_{\theta r}} g^{1+\delta_0}\Big)^{\frac{1}{1+\delta_0}} \le C_\theta\Big(G(r)+b(r)\Big),$$

which is (3.5) upon recalling $D(r) = r \fint_{Q_r} |\nabla u|^2$.

3.4 Local $L^{3+\delta}$ gain for u

Lemma 3.5 (Local $L^{3+\delta}$)**.** *There exist $\delta = \delta(\theta) \in (0, \frac{1}{3}]$ and $C_\theta < \infty$ such that*

$$\boxed{\Big(\fint_{Q_{\theta r}} |u|^{3+\delta}\Big)^{\frac{1}{3+\delta}} \;\le\; C\,\Phi(2r)^{1/3} \;+\; C_\theta\Big(\Phi^{1/3}E + \Phi + \Psi + \mathrm{Tail}_\theta\Big)(r)^{1/3}.} \tag{3.6}$$

Proof. Let $\chi = \eta\varphi$ with $\chi \equiv 1$ on $Q_{\theta r}$, $\operatorname{spt}\chi \subset Q_r$. Applying the spatial Gagliardo–Nirenberg inequality at each time and integrating,

$$\iint_{Q_r} |\chi u|^{10/3} \;\lesssim\; \Big(\sup_{t\in I_r}\int_{B_r} |\chi u|^2\Big)^{2/3} \iint_{Q_r} |\nabla(\chi u)|^2.$$

The L^2–sup factor is $\lesssim |B_r|\,E(r)$. Expanding $\nabla(\chi u) = \chi\nabla u + u\nabla\chi$ and using (3.1) plus $|\nabla\chi| \lesssim [(1-\theta)r]^{-1}$ and the tail, we obtain (after dividing by $|Q_{\theta r}| \sim r^5$)

$$\Big(\fint_{Q_{\theta r}} |\chi u|^{10/3}\Big)^{3/10} \;\lesssim_\theta\; \Big(\Phi^{1/3}E + \Phi + \Psi + \mathrm{Tail}_\theta\Big)(r)^{1/3}.$$

Interpolating between L^3 and $L^{10/3}$ on $Q_{\theta r}$ gives (3.6), using $\|u\|_{L^3(Q_{\theta r})} \le \|u\|_{L^3(Q_{2r})}$ and $\Phi(2r)^{1/3} = (\fint_{Q_{2r}} |u|^3)^{1/3}$. □

[to Lemma 3.5] *Gradient of the cut-off field.* Expanding $\nabla(\chi u) = \chi\nabla u + u\nabla\chi$,

$$\iint_{Q_r} |\nabla(\chi u)|^2 \le 2\iint_{Q_r} \chi^2|\nabla u|^2 + 2\iint_{Q_r} |u|^2|\nabla\chi|^2 \lesssim_\theta r\,[\Phi^{1/3}E + \Phi + \Psi](r) + r\ \mathrm{Tail}_\theta(r),$$

using (3.1) and $|\nabla\chi| \lesssim_\theta r^{-1}$ with support in $Q_r \setminus Q_{\theta r}$ (absorbed by Tail_θ).

3.5 Critical Moser iteration $L^\infty_t L^q_x$ (endpoint $q \downarrow 3^+$)

Proposition 3.6 (Critical Moser, endpoint-safe)**.** *Let $0 < \theta < 1$. There exists $\widehat{\delta} = \widehat{\delta}(\theta) > 0$ such that for every $q \in (3, 3+\widehat{\delta}]$,*

$$\boxed{\sup_{t\in I_{\theta r}} \|u(t)\|_{L^q(B_{\theta r})} \;\le\; C\Big(\fint_{Q_{2r}} |u|^{3+\delta}\Big)^{\frac{1}{3+\delta}} \;+\; C_\theta\Big(\Phi^{1/3}E + \Psi^{2/3} + \mathrm{Tail}_\theta^{1/3}\Big)(r),} \tag{3.7}$$

*where $\delta > 0$ is the exponent from Lemma 3.5. The constants are **uniform** as $q \downarrow 3$.*

Proof with corrected commutator and exponents. Fix cutoffs $\eta(t)$ and $\varphi(x)$ supported in Q_r with $\chi := \eta\varphi \equiv 1$ on $Q_{\theta r}$, $|\partial_t\eta| + |\nabla\varphi|^2 + |\Delta\varphi| \lesssim_\theta r^{-2}$. Test the projected equation

$$\partial_t u - \Delta u + \mathbb{P}\nabla\cdot(u\otimes u) = 0$$

with $Z := \chi^2|u|^{q-2}u$. Set $f := |u|^{q/2}$; then $|\nabla f|^2 \simeq |u|^{q-2}|\nabla u|^2$. The standard Moser calculus and Cauchy–Young yield

$$\frac{d}{dt}\int \chi^2|u|^q \;+\; c\iint \eta^2\varphi^2\,|\nabla f|^2 \;\le\; C\iint |\nabla\chi|^2|u|^q \;-\; \iint (u\otimes u) : \nabla(\mathbb{P}Z). \tag{3.8}$$

Local and commutator splitting. Write

$$-(u\otimes u):\nabla(\mathbb{P}Z) \;=\; -(u\otimes u):\nabla Z-(u\otimes u):\nabla\big([\mathbb{P},\varphi^2]|u|^{q-2}u\big)-(u\otimes u):\nabla\big(\mathbb{P}(\eta^2{-}1)\varphi^2|u|^{q-2}u\big).$$

The first piece is local and is handled in the standard way, absorbing $\varepsilon\iint \eta^2\varphi^2|\nabla f|^2$ into the left-hand side and paying $C_\theta r^{-2}\iint|u|^q$ on the right. The third piece has support in the corona $Q_r\setminus Q_{\theta r}$ and is controlled by the tail:

$$\Big|\iint(u\otimes u):\nabla\big(\mathbb{P}(\eta^2{-}1)\varphi^2|u|^{q-2}u\big)\Big| \;\lesssim_\theta\; r^{-1}\iint_{Q_r\setminus Q_{\theta r}}\Big(|u|^q{+}|\widetilde{p}|^{3/2}\Big) \;\lesssim_\theta\; \mathrm{Tail}_\theta(r)^{1/3}\Big(\iint_{Q_r}|u|^q\Big)^{2/3}.$$

It remains to bound the *commutator* piece.

Sharp commutator bound (order -1; HLS/CZ). For any tensor field F,

$$[\mathbb{P},\varphi^2]F \;=\; -\nabla(-\Delta)^{-1}\big(\nabla(\varphi^2)\cdot F\big) \;=\; I_1\big((\nabla\varphi^2)\cdot F\big),$$

where I_1 is the Riesz potential of order 1 in $\mathbb{R}^3$. By Hardy–Littlewood–Sobolev,

$$\|[\mathbb{P},\varphi^2]F\|_{L^q(\mathbb{R}^3)} \;\lesssim\; \|\nabla\varphi\|_\infty\,\|F\|_{L^p(\mathbb{R}^3)},\quad \frac{1}{q}=\frac{1}{p}-\frac{1}{3},\quad \tfrac{3}{2}<p<3,$$

with a constant independent of q as $q\downarrow 3$. We shall apply this with $F:=|u|^{q-2}u$.

Synchronizing exponents with the $L^{3+\delta}$ gain. Lemma 3.5 yields $u\in L^{3+\delta}(Q_{\theta r})$. For any $q\in(3,3+\widehat{\delta}]$ (to be chosen), Hölder gives

$$\||u|^{q-1}\|_{L^p(Q_{\theta r})}=\|u\|^{q-1}_{L^{(q-1)p}(Q_{\theta r})},\qquad (q-1)p=3+\delta \quad\Longleftrightarrow\quad p=\frac{3+\delta}{q-1}.$$

Choose q so that $3<q<3+\frac{2}{3}\,\delta$. Then $p=\frac{3+\delta}{q-1}\in(\frac{3}{2},3)$ and, crucially,

$$\frac{1}{q}=\frac{1}{p}-\frac{1}{3},$$

so the commutator mapping applies. Consequently,

$$\|[\mathbb{P},\varphi^2](|u|^{q-2}u)\|_{L^q(Q_{\theta r})} \;\lesssim\; \|\nabla\varphi\|_\infty\,\||u|^{q-1}\|_{L^p(Q_{\theta r})} \;\lesssim_\theta\; r^{-1}\,\|u\|^{q-1}_{L^{3+\delta}(Q_{\theta r})}. \tag{3.9}$$

Note that the implied constants are **uniform** as $q\downarrow 3$ (equivalently $p\downarrow 3/2$) because HLS constants depend smoothly on p,q in compact subintervals away from endpoints.

Closing the Moser inequality. Insert (3.9) in (3.8), absorb $\varepsilon\iint\eta^2\varphi^2|\nabla f|^2$ into the left, and bound the remaining terms by

$$C_\theta r^{-2}\iint_{Q_r}|u|^q \;+\; C_\theta r^{-q}\,\|u\|^{q-1}_{L^{3+\delta}(Q_r)}\,\|u\|_{L^{3+\delta}(Q_r)} \;+\; C_\theta\;\mathrm{Tail}_\theta(r)^{1/3}\Big(\iint_{Q_r}|u|^q\Big)^{2/3}.$$

Use Hölder in space–time and Lemma 3.5 to control both the $r^{-2}\iint|u|^q$ term and the mixed product by

$$C\Big(\fint_{Q_{2r}}|u|^{3+\delta}\Big)^{\frac{q}{3+\delta}} \;+\; C_\theta\Big(\Phi^{1/3}E+\Phi+\Psi+\mathrm{Tail}_\theta\Big)(r)^{\frac{q}{3+\delta}}.$$

Finally, take the essential supremum over $t\in I_{\theta r}$ and the q-th root; since $q\le 3+\widehat{\delta}$ with $\widehat{\delta}\le\frac{2}{3}\delta$, the powers remain uniformly controlled as $q\downarrow 3$, yielding (3.7). □

Remark 3.7 (Uniformity at the endpoint and admissible range)**.** The commutator uses the order--1 mapping $I_1:L^p\to L^q$ with $\frac{1}{q}=\frac{1}{p}-\frac{1}{3}$. Starting from $u\in L^{3+\delta}$ (Lemma 3.5), setting $p=\frac{3+\delta}{q-1}$ enforces

$$\tfrac{3}{2}<p<3 \quad\Longleftrightarrow\quad 3<q<3+\tfrac{2}{3}\,\delta,$$

so q may be taken arbitrarily close to 3 while keeping constants independent of $q\downarrow 3$. This corrects the endpoint issue in (B.7) and justifies the claim "uniform in $q\downarrow 3^+$".

3.6 ε–regularity at one scale

Theorem 3.8 (One–scale ε–regularity with explicit pressure tail)**.** *There exists a universal constant $\varepsilon_* > 0$ and a universal $C < \infty$ such that the following holds. Let $z_0 = (x_0, t_0)$ and $r > 0$, and normalize the pressure at scale $2r$ by*

$$\widetilde{p}(\cdot,t) := p(\cdot,t) - \fint_{B_{2r}(x_0)} p(\cdot,t)\,dx.$$

If

$$\Phi(2r) \;+\; \mathrm{Tail}_{1/2}(2r) \;\le\; \varepsilon_*, \tag{3.10}$$

where

$$\mathrm{Tail}_{1/2}(2r) \;:=\; \frac{1}{(2r)^2}\iint_{Q_{2r}(z_0)\setminus Q_r(z_0)} \Big(|u|^3 + |\widetilde{p}|^{3/2}\Big),$$

then u is smooth in $Q_{r/2}(z_0)$.

Proof. Fix two geometrical parameters $\theta_1 := \frac{3}{4}$ and $\theta_2 := \frac{1}{2}$. All constants below are universal (they may depend on θ_1, θ_2 only).

Step 1 (pressure and energy defects reduced to $\Phi(2r)$ + explicit tail). By (E1) applied at the *larger* scale $2r$ and by the elementary contraction of space–time averages,

$$\Psi(r) \;=\; \frac{1}{r^2}\iint_{Q_r} |\widetilde{p}|^{3/2} \;\le\; 4\,\frac{1}{(2r)^2}\iint_{Q_{2r}} |\widetilde{p}|^{3/2} \;\lesssim\; \Phi(2r) \;+\; \mathrm{Tail}_{1/2}(2r).$$

(Here the tail is precisely the annular budget on $Q_{2r}\setminus Q_r$ with pressure normalized at scale $2r$; it captures the harmonic contribution generated by the exterior.) Similarly, testing the local energy inequality on Q_{2r} and using the inter–scale Caccioppoli from Lemma 3.1 at $(R,\theta) = (2r, \frac{1}{2})$ yields

$$E(r) \;\lesssim\; \Phi(2r)^{2/3} \;+\; \Phi(2r) \;+\; \mathrm{Tail}_{1/2}(2r). \tag{3.11}$$

Step 2 (local $L^{3+\delta}$ with tail). Apply Lemma 3.5 at radius $R = 2r$ (with $\theta = \theta_1$) and use (3.11) and the bound for $\Psi(r)$ above to obtain, for some $\delta \in (0, \frac{1}{3}]$,

$$\Big(\fint_{Q_{\theta_1 r}} |u|^{3+\delta}\Big)^{\frac{1}{3+\delta}} \;\lesssim\; \Phi(2r)^{1/3} \;+\; \mathrm{Tail}_{1/2}(2r)^{1/3}. \tag{3.12}$$

Step 3 (critical Moser to $L^\infty_t L^q_x$). By the critical Moser Proposition 3.6, uniformly for any $q \in (3, 3+\widehat{\delta}]$,

$$\sup_{t\in I_{\theta_2 r}} \|u(t)\|_{L^q(B_{\theta_2 r})} \;\lesssim\; \Big(\fint_{Q_{\theta_1 r}} |u|^{3+\delta}\Big)^{\frac{1}{3+\delta}} \;+\; \Big(\Phi^{1/3}E + \Psi^{2/3} + \mathrm{Tail}_{\theta_2}^{1/3}\Big)(2r).$$

Using (3.11), the bound for $\Psi(r)$, and (3.12), the right-hand side is controlled by

$$C\Big(\Phi(2r)^{1/3} + \mathrm{Tail}_{1/2}(2r)^{1/3}\Big).$$

Thus,

$$\sup_{t\in I_{\theta_2 r}} \|u(t)\|_{L^q(B_{\theta_2 r})} \;\lesssim\; \Phi(2r)^{1/3} \;+\; \mathrm{Tail}_{1/2}(2r)^{1/3}. \tag{3.13}$$

Step 4 (Serrin–Prodi trigger and conclusion). Pick any $q > 3$ as above. Since $2/\infty + 3/q < 1$, the local Serrin–Prodi criterion (E3) applies on $Q_{\theta_2 r}$, giving smoothness on a slightly smaller cylinder, e.g. $Q_{r/2}$, provided the right-hand side of (3.13) is sufficiently small. This is ensured by the smallness hypothesis (3.10) after choosing ε_* small enough (universally). The conclusion follows. □

4 Good–λ descent and activation

Geometry, averages, normalized pressure. Fix $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and $r > 0$. Set

$$B_r := B_r(x_0), \qquad I_r := (t_0 - r^2, t_0], \qquad Q_r := B_r \times I_r, \qquad \fint_E f := \frac{1}{|E|} \int_E f.$$

On Q_r we use the *normalized pressure*

$$\widetilde{p}(\cdot, t) := p(\cdot, t) - \fint_{B_r} p(\cdot, t)\, dx.$$

Scale–invariant package. All quantities below are invariant under the parabolic scaling $u_\lambda(x,t) = \lambda\, u(x_0 + \lambda x, t_0 + \lambda^2 t)$, $p_\lambda = \lambda^2 p$, with $r \mapsto r/\lambda$:

$$\Phi(r) := \frac{1}{r^2} \iint_{Q_r} |u|^3, \qquad \Psi(r) := \frac{1}{r^2} \iint_{Q_r} |\widetilde{p}|^{3/2},$$
$$E(r) := \sup_{t \in I_r} \fint_{B_r} |u(\cdot, t)|^2\, dx, \qquad D(r) := \frac{1}{r} \iint_{Q_r} |\nabla u|^2.$$

We write $F(r) := D(r)$. For $\theta \in (0,1)$ the outer tail

$$\mathrm{Tail}_\theta(r) := \frac{1}{r^2} \iint_{Q_r \setminus Q_{\theta r}} \Big(|u|^3 + |\widetilde{p}|^{3/2} \Big)$$

measures the cost of localization; note $\mathrm{Tail}_\theta(r) \lesssim_\theta \Phi(r) + \Psi(r)$ and $\mathrm{Tail}_\theta(r) \downarrow 0$ as $\theta \uparrow 1$.

Finally, let

$$\rho := \frac{1}{k + |u|} \in (0, 1], \qquad \mathrm{Ent}(s) := \fint_{Q_s} |\nabla \rho|^2 \quad \text{(logarithmic entropy; cf. Lemma 3.2).}$$

Define the *defect functional*

$$\mathcal{M}(r) := F(r) + \lambda\, \mathrm{Ent}(r), \qquad \lambda \in (0, \lambda_0(\theta)].$$

4.1 Energy controlled by Φ, Ψ

Lemma 4.1 (Energy from Φ, Ψ)**.** *There exists a universal $C < \infty$ such that for all $r > 0$ (with the pressure normalized at scale $2r$)*

$$\boxed{E(r) \;\le\; C\Big(\Phi(2r)^{2/3} + \Phi(2r) + \Psi(2r) \Big).} \tag{4.1}$$

Proof. By scaling it suffices to work at $r = 1$. Take $\phi(x,t) = \psi(t)\varphi(x)^2$ with $\varphi \equiv 1$ on B_1, $\operatorname{supp}\varphi \subset B_2$, $|\nabla\varphi| + |\Delta\varphi| \lesssim 1$, $\psi \equiv 1$ on $(-1, 0]$, $|\psi'| \lesssim 1$. The local energy inequality (LEI) gives

$$\sup_{t \in (-1,0]} \int_{B_1} |u|^2 \;\lesssim\; \iint_{Q_2} |u|^2 (|\partial_t \phi| + |\Delta \phi|) \;+\; \iint_{Q_2} (|u|^2 + 2\widetilde{p})\, u \cdot \nabla \phi =: I_1 + I_2.$$

Hölder and scale invariance yield $I_1 \lesssim |Q_2|^{1/3} \big(\iint_{Q_2} |u|^3 \big)^{2/3} \sim \Phi(2)^{2/3}$. For the convective part in I_2, $|\nabla\phi| \lesssim 1$ gives $\iint_{Q_2} |u|^3 \sim \Phi(2)$. For the pressure contribution, use $2|\widetilde{p}||u||\nabla\phi| \le |\widetilde{p}|^{3/2} + |u|^3$ to get $\iint_{Q_2} |\widetilde{p}|^{3/2} \sim \Psi(2)$. Rescale back to r and divide by $|B_r|$ to match $E(r) = \sup_{t \in I_r} \fint_{B_r} |u|^2$. $\square$

4.2 Discrete monotonicity of $\mathcal{M}$ across scales

Proposition 4.2 (Monotonicity of $\mathcal{M}$)**.** *For each fixed $\theta \in (0,1)$ there exist $\kappa(\theta) \in (0,1)$, $C_\theta < \infty$ and $\lambda_0(\theta) > 0$ such that, for all $r > 0$ and all $\lambda \in (0, \lambda_0(\theta)]$,*

$$\boxed{\mathcal{M}(\theta r) \;\le\; \kappa(\theta)\,\mathcal{M}(r) \;+\; C_\theta\Big(\Phi(r)^{1/3}E(r) + \Phi(r) + \Psi(r)\Big).} \tag{4.2}$$

Proof. By the inter–scale Caccioppoli (Lemma 2.1 in §3) and the algebraic hole-filling,

$$F(\theta r) \;\le\; C_\theta\Big(\Phi^{1/3}E + \Phi + \Psi\Big)(r).$$

By the logarithmic Caccioppoli with hole–filling (Lemma 3.2),

$$\mathrm{Ent}(\theta r) \;\le\; \kappa_E(\theta)\,\mathrm{Ent}(r) \;+\; C_\theta\Big(\Phi^{1/3}E + \Phi + \Psi\Big)(r).$$

Add λ times the second inequality to the first and choose $\lambda \le \lambda_0(\theta)$ small so that $\kappa(\theta) := \max\{\kappa_E(\theta),\, 1 - \frac{\lambda}{2}\} < 1$; note that $\mathcal{M}(r) \ge \lambda\,\mathrm{Ent}(r)$. This gives (7.2). □

4.3 Bridge $M \to \Phi$

Lemma 4.3 (Bridge)**.** *For every $\theta \in (0,1)$ there exists $C_\theta < \infty$ such that, for all $r > 0$,*

$$\Phi(\theta r) \;\le\; C_\theta\big(\Phi(r) + \Psi(r)\big) \;+\; C_\theta\,M(r)^{3/2} \;+\; C_\theta\big(\Phi(r) + \Psi(r)\big)^{3/2}. \tag{C.3}$$

Here Φ, Ψ, E, F are the scale-invariant functionals of §B, $F(r) = D(r)$, and $M(r) = F(r) + \lambda\,\mathrm{Ent}(r)$ as in §C.

Proof. Fix $\theta \in (0,1)$ and $r > 0$. For a.e. t decompose

$$u(\cdot,t) = a(t) + w(\cdot,t), \qquad a(t) := \fint_{B_r} u(\cdot,t), \qquad \fint_{B_r} w(\cdot,t) = 0.$$

By Jensen and the mean–zero Poincaré–Gagliardo–Nirenberg on B_r,

$$\|u(t)\|^3_{L^3(B_{\theta r})} \;\le\; C\,\|w(t)\|^3_{L^3(B_{\theta r})} + C\int_{B_r} |u(x,t)|^3 dx, \qquad \|w(t)\|_{L^3(B_{\theta r})} \;\le\; C\,r^{1/2}\|\nabla u(t)\|_{L^2(B_r)}. \tag{4.3}$$

Combining these,

$$\|u(t)\|^3_{L^3(B_{\theta r})} \;\le\; C\,r^{3/2}\,\|\nabla u(t)\|^3_{L^2(B_r)} \;+\; C\int_{B_r} |u(x,t)|^3 dx. \tag{4.4}$$

Step 1 (Integrate in time and isolate $F^{3/2}$). Integrate (4.4) on $I_{\theta r}$ and use Hölder in time with $f(t) := \|\nabla u(t)\|_{L^2(B_r)}$:

$$\int_{I_{\theta r}} f(t)^3\,dt \;\le\; |I_{\theta r}|^{-1/2}\Big(\int_{I_{\theta r}} f(t)^2\,dt\Big)^{3/2} \;\le\; c_\theta\,r^{-1}\Big(\iint_{Q_r} |\nabla u|^2\Big)^{3/2}.$$

Hence

$$\iint_{Q_{\theta r}} |u|^3 \;\le\; C_\theta\,r^{3/2}\cdot r^{-1}\Big(\iint_{Q_r} |\nabla u|^2\Big)^{3/2} + C\iint_{Q_r} |u|^3 = C_\theta\,r^{1/2}\Big(\iint_{Q_r} |\nabla u|^2\Big)^{3/2} + C\iint_{Q_r} |u|^3.$$

Divide by $(\theta r)^2 \sim_\theta r^2$ and use $F(r) = \frac{1}{r}\iint_{Q_r} |\nabla u|^2$ to get the dimensionless form

$$\Phi(\theta r) \;\le\; C_\theta\,F(r)^{3/2} \;+\; C_\theta\,\Phi(r). \tag{4.5}$$

Step 2 (Linear pressure piece via normalized/anchored gauge). If one inserts a cutoff $\chi = \eta\phi$ with $\chi \equiv 1$ on $Q_{\theta r}$ and support in Q_r, every derivative of χ is supported on the annulus $A_r := Q_r \setminus Q_{\theta r}$. The pressure contribution in the LEI,

$$2 \iint p\, u \cdot \nabla(\chi^2)\,,$$

is handled using the *normalized pressure* $\tilde{p} := p - \fint_{B_r} p(\cdot,t)$ and the local Calderón–Zygmund bound (E1): this yields a linear contribution bounded by $C_\theta(\Phi(r)+\Psi(r))$, plus an outer corona tail. Working instead with the *anchored gauge* $p^b := p - \fint_{B_{2r_0}} p(\cdot,t)$ from §C.4 relocates the tail entirely into the explicit corona budget $\Xi_R(A_r)$, which by the packing estimate is absorbed into $C_\theta(\Phi(r)+\Psi(r))$. In either normalization we can strengthen (4.5) to

$$\Phi(\theta r) \;\le\; C_\theta\, F(r)^{3/2} \;+\; C_\theta\big(\Phi(r)+\Psi(r)\big). \tag{4.6}$$

Step 3 (Upgrade $F^{3/2}$ to $M^{3/2}$ and account for E). Since $M(r) = F(r) + \lambda\,\mathrm{Ent}(r) \ge F(r)$, we have $F(r)^{3/2} \le M(r)^{3/2}$. If one prefers to route through the instantaneous energy, the refined Young inequality gives

$$F(r)^{3/2} \;\le\; \varepsilon\big(F(r)+E(r)\big) \;+\; C_\varepsilon\big(F(r)+E(r)\big)^{3/2}.$$

Using Lemma C.1 (energy from Φ,Ψ) we have $E(r) \lesssim \Phi(2r)^{2/3} + \Phi(2r) + \Psi(2r) \lesssim \Phi(r)+\Psi(r)$ (harmless change of θ). Inserting this into the previous display yields an additional contribution of order $(\Phi(r)+\Psi(r))$ and $(\Phi(r)+\Psi(r))^{3/2}$, absorbed in the right-hand side. Combining with (4.6) we obtain

$$\Phi(\theta r) \;\le\; C_\theta\, M(r)^{3/2} \;+\; C_\theta\big(\Phi(r)+\Psi(r)\big) \;+\; C_\theta\big(\Phi(r)+\Psi(r)\big)^{3/2},$$

which is exactly (C.3). All constants depend only on θ, and every term is scale-invariant. □

Remark 4.4 (Normalization bookkeeping)**.** The proof keeps $F \equiv D$ (no switching of symbols), uses the pressure gauge either normalized at scale r or anchored at the fixed outer scale $R = 2r_0$ of §C.4, and records all cutoff/annular costs as a linear $(\Phi+\Psi)$ term (plus a corona tail absorbed by the anchored budget). No residual r factors appear after dividing by r^2 in the passage from $\iint |u|^3$ to $\Phi(\cdot)$; the factor $F(r)^{3/2}$ is exposed by

$$\int_{I_{\theta r}} \|\nabla u(\cdot,t)\|_{L^2(B_r)}^3\,dt \;\le\; |I_{\theta r}|^{-1/2}\Big(\iint_{Q_r} |\nabla u|^2\Big)^{3/2} \;=\; c_\theta\, r^{-1}\big(rF(r)\big)^{3/2} = c_\theta\, F(r)^{3/2}.$$

4.4 Anchored gauge, corona budget, and one–step inequality

For a fixed base scale $r_0 > 0$ set $R := 2r_0$ and define the *anchored pressure gauge*

$$\widehat{p}(x,t) := p(x,t) - \fint_{B_R} p(\cdot,t)\,dx.$$

For any measurable $E \subset Q_R$ introduce the budget

$$\Xi_R(E) := \frac{1}{r_0^2} \iint_E \Big(|u|^3 + |\widehat{p}|^{3/2}\Big).$$

For $r \le r_0$, $\Xi_R(Q_r) \simeq \Phi(r)+\Psi(r)$ (universal comparability). Denote the annulus $A_r := Q_r \setminus Q_{\theta r}$; the cutoffs used above have derivatives supported in A_r.

Lemma 4.5 (One–step inequality with explicit corona budget)**.** *There exist $\kappa \in (0,1)$ and $C < \infty$ (depending only on θ) such that, for all $r \le r_0$,*

$$\boxed{\mathcal{M}(\theta r) \;\le\; \kappa\,\mathcal{M}(r) \;+\; C\,\Xi_R(A_r).} \tag{4.7}$$

Proof. Revisit the proofs of Lemma 2.1 (Caccioppoli) and Lemma 3.2 (log–Caccioppoli). All derivatives of the cutoffs live on A_r; the only terms not absorbed into the contracting part (i.e., not contributing to $\kappa\,\mathcal{M}(r)$) are supported on A_r and are bounded by $r^{-2}\iint_{A_r}(|u|^3 + |\widetilde{p}|^{3/2})$. Replacing $\widetilde{p}$ by $\widehat{p}$ (anchored at scale R) changes the normalization by a universal constant. This gives (4.7). □

Packing of annuli (discrete Carleson bound). For the dyadic radii $r_j := \theta^j r_0$ the annuli A_{r_j} are pairwise disjoint in space–time; hence

$$\sum_{j=0}^{J-1} \Xi_R(A_{r_j}) \;\le\; C\,\Xi_R(Q_{r_0}) \;\lesssim\; \Phi(2r_0) + \Psi(2r_0). \tag{4.8}$$

4.5 Geometric descent and activation of ε–regularity

Theorem 4.6 (Good–λ at finite scales)**.** *Fix $\theta \in (0,1)$. There exist universal $c \in (0,1)$ and $C_0 < \infty$ such that, for $r_J = \theta^J r_0$,*

$$\boxed{\mathcal{M}(r_J) \;\le\; (1-c)^J\,\mathcal{M}(r_0) \;+\; C_0\big(\Phi(2r_0) + \Psi(2r_0)\big).} \tag{4.9}$$

In particular, if $\Phi(2r_0) + \Psi(2r_0) \le \varepsilon_1$ (universal), then there exists $j \le J_* = J_*(\mathcal{M}(r_0))$ such that $\Phi(2r_{j+1}) \le \varepsilon_*$ (universal), and the one–scale ε–regularity (Theorem 2.6) yields smoothness in $Q_{r_{j+1}/2}$.

Proof. Iterate (4.7):

$$\mathcal{M}(r_J) \;\le\; \kappa^J \mathcal{M}(r_0) \;+\; C\sum_{j=0}^{J-1} \kappa^{J-1-j}\,\Xi_R(A_{r_j}).$$

Choose any $c \in (0, 1-\kappa)$ and use $\kappa^{J-1-j} \le (1-c)^{J-1-j}$; then sum and apply (5.7) to obtain (4.9). If the exterior budget is small, $\Phi(2r_0) + \Psi(2r_0) \le \varepsilon_1$, then for J minimal with $(1-c)^J\mathcal{M}(r_0) \le \frac{1}{2}A_{\mathrm{crit}}$ (the threshold from Lemma 4.3) one gets $\mathcal{M}(r_J) \le A_{\mathrm{crit}}$. Applying the bridge (7.3) at r_J gives $\Phi(2r_{J+1}) \le C_\theta\varepsilon_1 + C_\theta A_{\mathrm{crit}}^{3/2}$ which we rename ε_* (universal after fixing θ). The one–scale criterion (Theorem 2.6) then applies. □

Remark (dichotomy). Without small exterior budget one still has the geometric decay (4.9), but the bridge (7.3) carries a linear $\Phi + \Psi$ piece, so smallness of Φ is not automatic at every step. This is the precise, non–misleading statement of the good–λ mechanism.

4.6 Maximal–function good–λ (distributional form)

Let $\mathcal{M}^*$ denote the parabolic Hardy–Littlewood maximal operator over cylinders centered at a point. For a fixed $\beta > 0$ define $Z(r) := \mathcal{M}(r) + \beta(\Phi + \Psi)(r)$.

Lemma 4.7 (Level–set inequality)**.** *There exist $A > 1$, $\lambda_0 > 0$, $\gamma \in (0,1)$, $C < \infty$ (depending only on θ) such that for all $\lambda \ge \lambda_0$,*

$$\boxed{\big|\{Q_r :\; \mathcal{M}^*Z > A\lambda\}\big| \;\le\; \gamma\,\big|\{Q_{2r} :\; \mathcal{M}^*Z > \lambda\}\big| \;+\; C\,\big|\{Q_{2r} :\; \mathcal{M}^*(\Phi+\Psi) > \lambda\}\big|.} \tag{4.10}$$

Sketch. Perform a parabolic Vitali selection on the superlevel set of $\mathcal{M}^*Z$ at height $A\lambda$. On each selected cylinder apply the one–step inequality (4.7). The contraction factor contributes to the first term on the right–hand side; all boundary/cutoff terms are supported on coronas and are absorbed by $\mathcal{M}^*(\Phi+\Psi)$. Summing over disjoint cylinders proves (4.10). □

Iterating (4.10) yields the same activation conclusion as Theorem 4.6 in the distribution–function language.

Editorial invariance and non–circularity. Every inequality above is adimensional and invariant under the NSE scaling. The pressure's nonlocality is tracked explicitly via the anchored gauge and the corona budget; no term is silently discarded. The defect $\mathcal{M}$ contracts before any regularity is invoked; the bridge is invoked only after $\mathcal{M}$ is small; the one–scale ε–regularity is the final local trigger. No step uses ESC–type inputs (ESS) to verify their own hypotheses. "'

5 Dynamic compactness and ancient profile

Fix a base point $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and a radius $r > 0$. We set

$$B_r := B_r(x_0), \qquad I_r := (t_0 - r^2, t_0], \qquad Q_r := B_r \times I_r.$$

Space–time averages are denoted by $⨍_E f := |E|^{-1} \int_E f$. The pressure gauge on Q_r is the time-dependent normalization

$$\widetilde{p}(\cdot,t) := p(\cdot,t) - ⨍_{B_r} p(\cdot,t)\,dx.$$

All dimensionless functionals are the ones fixed in Sections 3–4:

$$\Phi(r) := \frac{1}{r^2}\iint_{Q_r} |u|^3, \qquad \Psi(r) := \frac{1}{r^2}\iint_{Q_r} |\widetilde{p}|^{3/2}, \qquad E(r) := \sup_{t\in I_r} ⨍_{B_r} |u|^2,$$

$$F(r) := \frac{1}{r}\iint_{Q_r} |\nabla u|^2, \qquad \mathrm{Ent}(r) := ⨍_{Q_r} |\nabla \rho|^2 \quad \text{with} \quad \rho = \frac{1}{k+|u|},$$

and, for $\theta \in (0,1)$, the outer tail

$$\mathrm{Tail}_\theta(r) := \frac{1}{r^2}\iint_{Q_r \setminus Q_{\theta r}} \big(|u|^3 + |\widetilde{p}|^{3/2}\big) \lesssim_\theta \Phi(r) + \Psi(r), \qquad \mathrm{Tail}_\theta(r) \downarrow 0 \text{ as } \theta \uparrow 1.$$

All quantities are invariant under the parabolic scaling $u_\lambda(x,t) = \lambda u(x_0 + \lambda x, t_0 + \lambda^2 t)$, $p_\lambda = \lambda^2 p$, $r \mapsto r/\lambda$.

5.1 Canonical gauge and APMS control

For each $t < 0$ we fix a *canonical center* $x(t)$ (e.g. an L^3-median or center of mass) and define the *median radius* $\lambda(t) > 0$ as the unique number such that

$$\int_{B_{\lambda(t)}(x(t))} |u(\cdot,t)|^3\,dx \;=\; \tfrac{1}{2}\int_{\mathbb{R}^3} |u(\cdot,t)|^3\,dx. \tag{5.1}$$

We set the centered, normalized view

$$\widehat{u}(t,y) := \lambda(t)\,u\big(x(t) + \lambda(t)y, t\big),$$

so that $\|\widehat{u}(t)\|_{L^3(\mathbb{R}^3)} = \|u(t)\|_{L^3(\mathbb{R}^3)}$ for every t.

Lemma 5.1 (Gauge stability)**.** *There exists $C \geq 1$ such that, for $|h| \leq \lambda(t)^2$,*

$$|x(t+h) - x(t)| \;\leq\; C\,\lambda(t), \qquad C^{-1} \leq \frac{\lambda(t+h)}{\lambda(t)} \leq C.$$

Proof. Fix t and h with $|h| \leq \lambda(t)^2$. Apply the terminal slicing Lemma 5.3 below on $Q_{\lambda(t)}(x(t), t)$ to estimate, for a.e. $s \in [t - \lambda(t)^2, t]$,

$$\|u(s)\|^3_{L^3(B_{\theta\lambda(t)}(x(t)))} \;\lesssim_\theta\; \frac{1}{\lambda(t)^2} \iint_{Q_{\lambda(t)}(x(t),t)} \big(|u|^3 + |\widetilde{p}|^{3/2}\big) + \lambda(t) \iint_{Q_{\lambda(t)}(x(t),t)} |\nabla u|^2.$$

By the discrete APMS control (proved in Lemma 5.2 below) the right-hand side is uniformly bounded (scale-invariant). Hence the L^3-mass inside $B_{\theta\lambda(t)}(x(t))$ is comparable, along the window $[t - \lambda(t)^2, t]$, to the total mass at time t. The median property (5.1) then forces both $x(\cdot)$ and $\lambda(\cdot)$ to vary by at most a fixed factor across time increments $|h| \leq \lambda(t)^2$ (a standard compactness/median-perturbation argument using that half-mass balls cannot drift more than their radius without violating the median equality). This yields the claim. □

For a geometric dyadic chain $r_j := \theta^j r_0$ $(0 < \theta < 1)$ anchored at $r_0 \sim \lambda(t)$ we define the *APMS (all-parabolic-multi-scales) dissipation profile*

$$F_j(t) := F(r_j; x(t), t) = \frac{1}{r_j} \iint_{Q_{r_j}(x(t),t)} |\nabla u|^2.$$

Lemma 5.2 (APMS Carleson budget)**.** *There exists $C_{\mathrm{APMS}} < \infty$ universal such that, for every $t < 0$,*

$$\sum_{j \geq 0} F_j(t) \;\leq\; C_{\mathrm{APMS}}. \tag{5.2}$$

Moreover, still at scale $r_0 \sim \lambda(t)$,

$$\Phi(r_0) + \Psi(r_0) \;\simeq\; \frac{1}{r_0^2} \iint_{Q_{r_0}(x(t),t)} \big(|u|^3 + |\widetilde{p}|^{3/2}\big), \qquad F(r_0) \;\lesssim\; \Phi(r_0)^{1/3} E(r_0) + \Phi(r_0) + \Psi(r_0).$$

Proof. Use the one-step inequality (good-λ step) from Proposition **??** together with Lemma 3.2 (log–Caccioppoli) to obtain, for cutoffs supported on annuli $A_r := Q_r \setminus Q_{\theta r}$,

$$\mathcal{M}(\theta r) \;\leq\; \kappa\,\mathcal{M}(r) + C\,\Xi_R(A_r), \qquad \Xi_R(E) := \frac{1}{r_0^2} \iint_E \big(|u|^3 + |\widetilde{p}|^{3/2}\big),$$

with $\kappa \in (0,1)$ universal and $R := 2r_0$. Iterating along $r_j = \theta^j r_0$ and summing, the disjointness of the annuli A_{r_j} in space–time yields

$$\sum_{j=0}^{J} \mathcal{M}(r_j) \;\leq\; C\Big(\mathcal{M}(r_0) + \Xi_R(Q_{r_0})\Big).$$

Since $F \leq \mathcal{M}$, (5.2) follows by letting $J \to \infty$. The two additional comparabilities are precisely the local CZ control of pressure (E1) and the inter-scale Caccioppoli from Lemma 3.1. □

5.2 Parabolic slicing: from cylinder averages to fixed time

Lemma 5.3 (Terminal slicing to a fixed time)**.** *Let $z_0 = (x_0, t_0)$, $r > 0$ and $\theta \in (0,1)$. For a.e. $t \in (t_0 - (\theta r)^2, t_0]$,*

$$\int_{B_{\theta r}(x_0)} |u(\cdot,t)|^3 \;\lesssim_\theta\; \frac{1}{r^2} \iint_{Q_r(z_0)} \Big(|u|^3 + |\widetilde{p}|^{3/2}\Big) \;+\; r \iint_{Q_r(z_0)} |\nabla u|^2. \tag{5.3}$$

The same estimate holds if $B_{\theta r}(x_0)$ is replaced by an annulus $B_r \setminus B_{\theta r}$, with the same right-hand side.

Proof. Fix space–time cutoffs $\varphi \in C_c^\infty(B_r)$ and $\eta \in C_c^\infty((t_0 - r^2, t_0])$ such that

$$\varphi \equiv 1 \text{ on } B_{\theta r}, \quad |\nabla\varphi| \lesssim [(1-\theta)r]^{-1}, \quad |\Delta\varphi| \lesssim [(1-\theta)r]^{-2}, \qquad \eta \equiv 1 \text{ on } [t_0-(\theta r)^2, t_0], \quad |\eta'| \lesssim r^{-2}.$$

Apply the LEI with $\phi(x,s) = \eta(s)\varphi(x)^2$ and the normalized pressure $\widetilde{p}$ on Q_r; then evaluate at time t (standard mollification in time justifies the evaluation). Using Hölder, Poincaré–Gagliardo–Nirenberg on B_r (for the zero-mean oscillation), and the bounds on the derivatives of ϕ, one finds

$$\int |u|^2\varphi^2(\cdot,t) \;+\; \iint \eta\,|\nabla u|^2\varphi^2 \;\lesssim\; \iint |u|^2(|\partial_t\phi| + |\Delta\phi|) \;+\; \iint \big(|u|^3 + |\widetilde{p}|^{3/2}\big)|\nabla\phi|.$$

Now invoke the L^3–L^2 local GN bound at time t,

$$\|u(t)\|_{L^3(B_{\theta r})}^3 \;\lesssim\; \|u(t)\|_{L^2(B_r)}^{3/2}\,\|\nabla u(t)\|_{L^2(B_r)}^{3/2},$$

and absorb the spatial gradient via Young with $\varepsilon > 0$ (fix ε small and use that $\|\nabla u\|_{L^2(Q_r)}^2$ appears on the left). The pressure part is handled through the CZ estimate for $\widetilde{p}$ on B_r. This yields (5.3). The annular case follows by replacing φ with an annular cutoff supported in $B_r \setminus B_{\theta r}$. □

5.3 Uniform L^3 tails (no time averaging)

Let $A_j := \{y \in \mathbb{R}^3 : 2^j \le |y| \le 2^{j+1}\}$ in the rescaled variable y (so the physical annulus has radii $\sim r_j := \lambda(t)2^j$). Consider an annular cutoff χ_j supported in a thickened annulus $A_j^\sharp$ with $|\nabla\chi_j| \lesssim r_j^{-1}$.

Proposition 5.4 (Uniform L^3 tails of the profile in the gauge)**.** *Let u_b be the field in the dynamic gauge of §5 (centers $x(t)$ and median radii $\lambda(t)$). For every $\varepsilon > 0$ there exists $R(\varepsilon) \ge 1$ such that*

$$\sup_{t<0} \int_{|y|>R(\varepsilon)} |u_b(t,y)|^3\,dy \le \varepsilon.$$

Proof. Step 0 (annuli and cylinders). In the rescaled variable y, set $A_j := \{2^j \le |y| \le 2^{j+1}\}$, $A_j^\sharp$ a fixed thickening of A_j, $r_j := 2^j$, and $I_j(t) := [\,t - r_j^2, t\,]$. Write $E_j(t) := I_j(t) \times A_j^\sharp$. Given $J \in \mathbb{N}$, we write

$$\int_{|y|>2^J} |u_b(t,y)|^3 = \sum_{j\ge J} \int_{A_j} |u_b(t,y)|^3\,dy.$$

Step 1 (terminal annular slicing). Let $\chi_j \in C_c^\infty(A_j^\sharp)$ with $\chi_j \equiv 1$ on A_j and $\|\nabla\chi_j\|_\infty \lesssim r_j^{-1}$. Applying the **terminal slicing lemma** (Lemma 5.3) with a space–time cutoff supported in $E_j(t)$ (the annular version is identical: one only replaces the ball by an annulus in the proof), we obtain for a.e. t:

$$\int_{A_j} |u_b(\cdot,t)|^3 \;\lesssim\; r_j^{-2} \iint_{E_j(t)} \big(|u_b|^3 + |\widetilde{p}|^{3/2}\big) \;+\; r_j \iint_{E_j(t)} |\nabla u_b|^2. \tag{5.4}$$

See Lemma 5.3 for the ball case; the same estimate with an annular cutoff (derivatives localized in $A_j^\sharp$) is proved with the same steps and dimensionless constants. turn0file13

Step 2 (annular Caccioppoli with $1/r$ gain). Return to the local energy inequality (LEI) with normalized pressure and prove an **annular Caccioppoli**:

$$r_j \iint_{E_j(t)} |\nabla u_b|^2 \;\lesssim\; r_j^{-1} \iint_{E_j^\star(t)} \big(|u_b|^3 + |\widetilde{p}|^{3/2}\big), \tag{5.5}$$

where $E_j^\star(t)$ is a thickened annulus containing $E_j(t)$ and still disjoint in j. The proof is standard: take $\varphi = \eta^2\chi_j^2$ in the LEI (with η temporal on $I_j(t)$, χ_j annular), normalize the pressure, and estimate each cutoff term using that $\nabla\chi_j$ lives in the corona with $\|\nabla\chi_j\|_\infty \lesssim r_j^{-1}$. The quadratic contributions of u_b are handled by Hölder and an annular Poincaré (via $|A_j^\sharp| \sim r_j^3$), and a fraction of $\iint |\nabla u_b|^2\chi_j^2$ is absorbed by the classical *hole filling* (Widman algebra). The final outcome is (5.5), completely dimensionless.

Step 3 (from slicing to a pure exterior budget). Inserting (5.5) into (5.4) we obtain

$$\int_{A_j} |u_b(\cdot,t)|^3 \;\lesssim\; r_j^{-2}\iint_{E_j(t)}\big(|u_b|^3 + |\tilde p|^{3/2}\big) \;+\; r_j^{-1}\iint_{E_j^\star(t)}\big(|u_b|^3 + |\tilde p|^{3/2}\big). \tag{5.6}$$

Step 4 (packing of annuli with anchored gauge). Fix a base scale $r_0 \sim \lambda(t)$ and the *anchored gauge* of §4.4. Define the budget

$$\Xi_R(E) := \frac{1}{r_0^2}\iint_E \big(|u_b|^3 + |\tilde p|^{3/2}\big).$$

For our sets $E_j(t)$ (and $E_j^\star(t)$), *the annuli are disjoint in space–time* and the **packing** of §4.4 gives

$$\sum_{j=J}^{\infty}\Xi_R\big(E_j(t)\big) \;+\; \sum_{j=J}^{\infty}\Xi_R\big(E_j^\star(t)\big) \;\lesssim\; \Xi_R\big(Q_{r_0}\big) \;\simeq\; \Phi(2r_0) + \Psi(2r_0), \tag{5.7}$$

uniformly in t (comparability via local CZ of the normalized pressure and dimensionless definitions). See §4.4 and (4.8). turn0file13

Step 5 (sum over $j \ge J$). Summing (5.6) in $j \ge J$ and using (5.7):

$$\sum_{j\ge J}\int_{A_j} |u_b(\cdot,t)|^3 \;\lesssim\; \sum_{j\ge J} r_j^{-2}\, r_0^2\, \Xi_R\big(E_j(t)\big) \;+\; \sum_{j\ge J} r_j^{-1}\, r_0^2\, \Xi_R\big(E_j^\star(t)\big).$$

Since $r_j/r_0 = 2^j$, the two series carry weights 2^{-2j} and 2^{-j}, respectively. By (5.7) we conclude

$$\sum_{j\ge J}\int_{A_j} |u_b(\cdot,t)|^3 \;\lesssim\; \left(\sum_{j\ge J} 2^{-2j} + \sum_{j\ge J} 2^{-j}\right)\big(\Phi(2r_0) + \Psi(2r_0)\big) \;\lesssim\; 2^{-J}\,\big(\Phi(2r_0) + \Psi(2r_0)\big).$$

Finally, by the choice of the median radius $\lambda(t)$ and the comparabilities of §5, $\Phi(2r_0) + \Psi(2r_0)$ is uniformly bounded in t. Given $\varepsilon > 0$, take J large so that the right-hand side bound is $\le \varepsilon$ and set $R(\varepsilon) = 2^J$. The supremum in t follows by uniformity of the constants. □

5.4 Spatial translations and Kolmogorov–Riesz

Lemma 5.5 (Uniform translation modulus)**.** *There exist $q > 3$, $\alpha \in (0,1)$, and for each $R > 0$ a constant $C(R)$ such that, for all $|h| \le 1$ and all $t < 0$,*

$$\|\tau_h\widehat u(t) - \widehat u(t)\|_{L^3(B_R)} \;\le\; C(R)\,|h|^\alpha, \tag{5.8}$$

where $\tau_h f(\cdot) = f(\cdot + h)$.

Proof. From the critical Moser Proposition 3.6 (applied in rescaled balls) there exists $q > 3$ with $\sup_{t<0}\|\widehat u(t)\|_{L^q(B_{R+1})} \le C(R)$. By the elementary translation estimate and Poincaré,

$$\|\tau_h f - f\|_{L^2(B_R)} \;\le\; |h|\,\|\nabla f\|_{L^2(B_{R+1})}.$$

Interpolate between L^2 and L^q: choose $\theta \in (0,1)$ with $\frac{1}{3} = \frac{\theta}{2} + \frac{1-\theta}{q}$. Then

$$\|\tau_h f - f\|_{L^3(B_R)} \;\le\; \|\tau_h f - f\|_{L^2(B_R)}^{\theta}\,\|\tau_h f - f\|_{L^q(B_R)}^{1-\theta} \;\le\; C(R)\,|h|^\theta \|\nabla f\|_{L^2(B_{R+1})}^{\theta}.$$

Apply this to $f = \widehat u(t)$ and use the inter-scale Caccioppoli (Lemma 3.1) to bound $\|\nabla\widehat u(t)\|_{L^2(B_{R+1})}$ uniformly in t. Set $\alpha := \theta$. □

Corollary 5.6 (Dynamic L^3 compactness)**.** *The set $\mathcal{K} := \{\widehat{u}(t) : t < 0\} \subset L^3(\mathbb{R}^3)$ is relatively compact.*

Proof. By Proposition **??** the L^3 tails are uniform in time; on each ball B_R, Lemma 5.5 provides a uniform translation modulus. Kolmogorov–Riesz implies precompactness in $L^3(B_R)$ for each R, and the tail control upgrades this to $L^3(\mathbb{R}^3)$. □

5.5 Extraction of an ancient profile; temporal modulus; global L^3 bound

Assume for contradiction that $t = 0$ is singular. Let $t_n \uparrow 0$, $x_n := x(t_n)$, $\lambda_n := \lambda(t_n)$ and define the dynamically centered rescalings

$$u^{(n)}(x,t) := \lambda_n\, u\big(x_n + \lambda_n x,\ t_n + \lambda_n^2 t\big), \qquad \widetilde{p}^{(n)}(x,t) := \lambda_n^2\, \widetilde{p}\big(x_n + \lambda_n x,\ t_n + \lambda_n^2 t\big).$$

If $\lambda_n \downarrow 0$, the time domains contain $(-T_n, 0]$ with $T_n \to \infty$; any limit is ancient.

Uniform local bounds and time modulus. Fix $R, S > 0$. By Sections 3–4 and Lemma 5.2 there exists $C_{R,S}$ independent of n such that

$$\iint_{(-S,0]\times B_{R+2}} |\nabla u^{(n)}|^2 \ \le\ C_{R,S}, \tag{5.9}$$

$$\sup_{t\in(-S,0]} \|u^{(n)}(t)\|_{L^q(B_{R+1})} \ +\ \iint_{(-S,0]\times B_{R+1}} |u^{(n)}|^{3+\delta} \ \le\ C_{R,S} \quad (q > 3,\ \delta > 0), \tag{5.10}$$

$$\|\widetilde{p}^{(n)}\|_{L^{3/2}((-S,0]\times B_{R+1})} \ +\ \sup_{t\in(-S,0]} \|u^{(n)}(t)\|_{L^2(B_{R+1})} \ \le\ C_{R,S}. \tag{5.11}$$

From the projected NSE,

$$\partial_t u^{(n)} = \Delta u^{(n)} - \mathbb{P}\nabla\cdot\,(u^{(n)} \otimes u^{(n)}),$$

we obtain (using (5.9) and (5.10))

$$\|\partial_t u^{(n)}\|_{L^{5/4}(-S,0;W^{-1,5/4}(B_R))} \ \le\ C_{R,S}. \tag{5.12}$$

Interpolating (5.12) with the $L^\infty_t L^q_x$ bound (5.10) yields a uniform time Hölder modulus in L^3 on B_R: there exists $\alpha = \alpha(q) \in (0,1)$ such that

$$\|u^{(n)}(t+\tau) - u^{(n)}(t)\|_{L^3(B_R)} \ \le\ C_{R,S}|\tau|^\alpha \qquad (t, t+\tau \in [-S, 0]). \tag{5.13}$$

Compactness and strong convergence. By Aubin–Lions (Simon) with (5.9)–(5.12), the sequence is relatively compact in $L^3((-S,0]\times B_R)$. Using a diagonal argument in R, S and the uniform tails from Proposition **??**, we extract a subsequence (not relabeled) and a limit

$$u^{(n)} \ \to\ U \qquad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0]). \tag{5.14}$$

Stability of the LEI under strong L^3_{loc} convergence and the CZ control for the pressure shows that U is a suitable ancient solution on $\mathbb{R}^3 \times (-\infty, 0]$.

Global L^3 bound and continuity at $t = 0$. The uniform tails pass to the limit; on balls, interpolation between L^2 and L^q limits gives, for every $S > 0$,

$$\sup_{t\in(-S,0]} \|U(t)\|_{L^3(\mathbb{R}^3)} \ \le\ C_S \ < \infty. \tag{5.15}$$

The time modulus (5.13) also passes to the limit, yielding local L^3-continuity in time. Tightness of tails then gives

$$U(t) \ \to\ U(0) \quad \text{in } L^3(\mathbb{R}^3) \ \text{ as } t \uparrow 0.$$

5.6 Non-vanishing of the ancient profile

Recall the one-scale ε-regularity (Theorem 3.8): there exists $\varepsilon_* > 0$ such that

$$\Phi(2r) \le \varepsilon_* \quad \Longrightarrow \quad u \text{ is smooth in } Q_{r/2}. \tag{5.16}$$

Lemma 5.7 (No-vanishing)**.** *Let $(x_*, 0)$ be a singular point of u. For any sequence $t_n \uparrow 0$ and their associated rescalings $u^{(n)}$ as above, for each fixed $r > 0$ we have*

$$\Phi_{u^{(n)}}(2r) \;=\; \Phi_u\big(2\,\lambda(t_n)\,r\big) \;>\; \varepsilon_* \qquad \textit{for all sufficiently large } n. \tag{5.17}$$

Along the subsequence extracted in (5.14) *we therefore have*

$$\Phi_U(2r) \;\ge\; \tfrac{1}{2}\,\varepsilon_* \qquad \forall\, r > 0, \tag{5.18}$$

in particular, $U \not\equiv 0$.

Proof. If there existed a sequence $r_k \downarrow 0$ with $\Phi_u(2r_k) \le \varepsilon_*$, then Theorem 3.8 would imply smoothness near $(x_*, 0)$, contradicting singularity. Hence $\Phi_u(2r) > \varepsilon_*$ for all sufficiently small r. By scale invariance of Φ and since $\lambda(t_n) \downarrow 0$,

$$\Phi_{u^{(n)}}(2r) \;=\; \Phi_u\big(2\,\lambda(t_n)\,r\big) \;>\; \varepsilon_*$$

for all n large enough, which proves (5.17). The strong L^3 convergence on Q_{2r} in (5.14) yields $\Phi_{u^{(n)}}(2r) \to \Phi_U(2r)$. Choose n large so that $\big|\Phi_{u^{(n)}}(2r) - \Phi_U(2r)\big| \le \varepsilon_*/2$; then (5.18) follows. □

Summary of Section 5. We constructed a dynamic gauge $(x(t), \lambda(t))$ whose stability across parabolic windows is quantified (Lemma 5.1); a discrete APMS budget (Lemma 5.2) controls the multi-scale dissipation. The terminal slicing Lemma 5.3 converts cylinder budgets into time-point bounds, and together with the APMS estimate yields uniform-in-time L^3 tails (Proposition **??**). A translation modulus (Lemma 5.5) and Kolmogorov–Riesz give compactness (Corollary 5.6); with Aubin–Lions we extract an ancient suitable limit U with a global critical L^3 bound and continuity at $t = 0$. Finally, the non-vanishing Lemma 5.7 shows that Φ_U stays uniformly away from zero at every scale, a fingerprint of blow-up that will contradict the boundary smoothness forced later by the ESS step.

6 Critical L^3 existence and stability; weak–strong identification; contradiction

Function spaces and Duhamel operator

Fix $0 < \delta < 1$ and $T \in (0, 1]$. We work on

$$Y_T \;:=\; L^\infty_t([-1, -1+T]; L^3_x) \;\cap\; L^{3+\delta}_{t,x}\big([-1, -1+T] \times \mathbb{R}^3\big).$$

For $f \in L^3(\mathbb{R}^3)$ and $t \in [-1, -1+T]$ consider the mild/Duhamel map

$$\mathcal{T}(u)(t) \;=\; \mathrm{e}^{(t+1)\Delta} f \;-\; \int_{-1}^{t} \mathrm{e}^{(t-s)\Delta}\,\mathbb{P}\,\nabla\cdot\big(u \otimes u\big)(s)\,ds. \tag{6.1}$$

We shall repeatedly use two standard ingredients.

(S) Semigroup with one spatial derivative. For $1 < p \le q \le \infty$ and $\tau > 0$,

$$\big\|\mathrm{e}^{\tau\Delta}\nabla\cdot F\big\|_{L^q_x} \;\le\; C\,\tau^{-\frac{1}{2}-\frac{3}{2}\left(\frac{1}{p}-\frac{1}{q}\right)}\,\|F\|_{L^p_x}, \qquad (\mathbb{P} \text{ bounded on } L^q). \tag{6.2}$$

(HLS) One-dimensional fractional integral in time. For the kernel $k(\tau) = \tau^{-(1-\alpha)}\mathbf{1}_{\tau>0}$ with $0 < \alpha < 1$ and $1 < m < \infty$, if $1/r = 1/m - \alpha > 0$ then

$$\Big\| \int_{-1}^{t} k(t-s)\, g(s)\, ds \Big\|_{L_t^r(-1,0)} \;\le\; C_{\alpha,m}\, \|g\|_{L_t^m(-1,0)}. \tag{6.3}$$

6.1 Bilinear time match (space-time critical gain)

Proposition 6.1 (Bilinear estimate)**.** *Let $F = u \otimes u$ and fix $p_x = \frac{3+\delta}{2}$. From (6.2) with $(p,q) = (p_x, 3)$ we have*

$$\big\| \mathrm{e}^{\tau\Delta}\, \mathbb{P}\nabla\cdot F \big\|_{L_x^3} \;\le\; C\, \tau^{-\beta}\, \|F\|_{L_x^{p_x}}, \qquad \beta \;=\; \frac{3}{3+\delta}. \tag{6.4}$$

Set $g(s) := \|u(s)\|_{L_x^{3+\delta}}^2 \in L_t^m$ with $m = \frac{3+\delta}{2}$, and put $\alpha := 1 - \beta = \frac{\delta}{3+\delta}$. Then

$$\Big\| \int_{-1}^{t} \mathrm{e}^{(t-s)\Delta} \mathbb{P}\nabla\cdot (u\otimes u)(s)\, ds \Big\|_{L_t^r L_x^3} \;\le\; C\, \|g\|_{L_t^m} \;=\; C\, \|u\|_{L_{t,x}^{3+\delta}}^2, \qquad \frac{1}{r} = \frac{1}{m} - \alpha = \frac{2-\delta}{3+\delta}. \tag{6.5}$$

Moreover, since $\beta < 1$,

$$\sup_{t\in[-1,-1+T]} \Big\| \int_{-1}^{t} \mathrm{e}^{(t-s)\Delta} \mathbb{P}\nabla\cdot (u\otimes u)(s)\, ds \Big\|_{L_x^3} \;\le\; C \Big(\int_0^T \tau^{-\beta}\, d\tau \Big) \|u\|_{L_{t,x}^{3+\delta}}^2 \;=\; C\, T^{\alpha}\, \|u\|_{L_{t,x}^{3+\delta}}^2. \tag{6.6}$$

Proof. The semigroup bound (6.2) with $(p,q) = (p_x, 3)$ yields (6.4) since

$$\frac{1}{2} + \frac{3}{2}\Big(\frac{1}{p_x} - \frac{1}{3}\Big) \;=\; \frac{1}{2} + \frac{3}{2}\Big(\frac{2}{3+\delta} - \frac{1}{3}\Big) \;=\; \frac{3}{3+\delta} \;=:\; \beta.$$

Applying (6.3) with kernel exponent $1 - \alpha = \beta$ and $g(s) = \|u(s)\|_{L^{3+\delta}}^2$ gives (6.5) with $1/r = 1/m - \alpha > 0$. The bound (6.6) follows by direct integration of $\tau^{-\beta}$ on $(0,T)$ since $\beta < 1$. □

6.2 Uniform-time L^3 existence and Lipschitz stability

Proposition 6.2 (Uniform L^3 time and Lipschitz stability)**.** *Let $\delta \in (0,1)$ and set $\alpha := \frac{\delta}{3+\delta} \in (0,1)$. For $f \in L^3(\mathbb{R}^3)$ and $\mathcal{T}$ as in (6.1):*

1. Linear bounds. $\|\mathrm{e}^{(t+1)\Delta} f\|_{L_t^\infty L_x^3} \le \|f\|_{L^3}$ *and* $\mathrm{e}^{(\cdot+1)\Delta} f \in C([-1,-1+T]; L^3)$. *Moreover*

$$\|\mathrm{e}^{(t+1)\Delta} f\|_{L_{t,x}^{3+\delta}([-1,-1+T]\times\mathbb{R}^3)} \;\le\; C\, T^{\frac{1-\delta}{2}}\, \|f\|_{L^3}. \tag{6.7}$$

2. Bilinear bounds. *With $p_x = \frac{3+\delta}{2}$ and α as above,*

$$\Big\| \int_{-1}^{t} \mathrm{e}^{(t-s)\Delta} \mathbb{P}\nabla\cdot (u\otimes u)(s)\, ds \Big\|_{L_t^{3+\delta} L_x^{3+\delta}} \le C\, \|u\|_{L_{t,x}^{3+\delta}}^2, \tag{6.8}$$

$$\Big\| \int_{-1}^{t} \mathrm{e}^{(t-s)\Delta} \mathbb{P}\nabla\cdot (u\otimes u)(s)\, ds \Big\|_{L_t^\infty L_x^3} \le C\, T^{\alpha}\, \|u\|_{L_{t,x}^{3+\delta}}^2. \tag{6.9}$$

3. Contraction on Y_T. *There is $C = C(\delta)$ such that for all $u, v \in Y_T$,*

$$\|\mathcal{T}(u)\|_{Y_T} \;\le\; \|f\|_{L^3} + C\, T^{\alpha}\, \|u\|_{Y_T}^2, \qquad \|\mathcal{T}(u) - \mathcal{T}(v)\|_{Y_T} \;\le\; C\, T^{\alpha}\, (\|u\|_{Y_T} + \|v\|_{Y_T})\, \|u - v\|_{Y_T}.$$

Hence, if $4C\, T^{\alpha} \|f\|_{L^3} < 1$, $\mathcal{T}$ is a contraction on the ball $\{u \in Y_T : \|u\|_{Y_T} \le 2\|f\|_{L^3}\}$, yielding a unique mild solution $u \in Y_T$ with

$$\|u\|_{L_t^\infty L_x^3([-1,-1+T])} \;\le\; 2\, \|f\|_{L^3}. \tag{6.10}$$

4. Uniform L^3 stability. *If $f_n \to f$ in L^3 and $\sup_n \|f_n\|_{L^3} \le M$, then there is $T_0 = T_0(M) > 0$ such that the corresponding mild solutions u_n, u on $[-1, -1+T_0]$ satisfy*

$$\sup_{t\in[-1,-1+T_0]} \|u_n(t) - u(t)\|_{L^3} \;\le\; 2\,\|f_n - f\|_{L^3}.$$

Proof. The linear bounds follow from L^3-contractivity of the heat semigroup and standard smoothing; (6.7) comes from $\|e^{\tau\Delta} f\|_{L^{3+\delta}} \lesssim \tau^{-\sigma}\|f\|_{L^3}$ with $\sigma = \frac{\delta}{2(3+\delta)}$, integrating $\tau^{-\sigma(3+\delta)} = \tau^{-\delta/2}$ on $[0,T]$. The bilinear estimates (6.8)–(6.9) are exactly (6.5)–(6.6) with the identification $L^r_t L^3_x \hookrightarrow L^{3+\delta}_{t,x}$ by Hölder on finite windows. The mapping bounds and contraction follow by standard fixed-point theory, using T^α small. Stability uses the same bilinear Lipschitz bound on the difference and contraction for $T_0(T_0(M))$. □

6.3 Weak–strong identification on short windows

Lemma 6.3 (Weak–strong uniqueness, short windows)**.** *Let $I = [t_0, t_0 + T]$. Suppose v is a mild solution on $\mathbb{R}^3 \times I$ with $v \in L^\infty_t L^3_x$ and $\nabla v \in L^2_{\mathrm{loc}}(I; L^2_{\mathrm{loc}})$, and u is a suitable solution on $\mathbb{R}^3 \times I$ with $u(t_0) = v(t_0) \in L^3$. If $C\,T^\alpha \|v(t_0)\|_{L^3} < 1$ (with $\alpha = \delta/(3+\delta)$), then $u \equiv v$ on I.*

Proof. Set $z = u - v$. Then $\partial_t z - \Delta z + \nabla q = -\nabla\cdot(u\otimes z + z\otimes v)$, $\nabla\cdot z = 0$, $z(t_0) = 0$. Fix a smooth spatial cutoff χ_R with $\chi_R \equiv 1$ on B_R and $\operatorname{supp}\chi_R \subset B_{2R}$. Testing the equation for z against $z\,\chi_R^2$, integrating in space-time, using the LEI for u and classical identities for v, and letting $R \to \infty$ (corona terms vanish by the uniform L^3 tails from §4), we obtain the energy inequality

$$\|z(t)\|_{L^2}^2 + \int_{t_0}^t \|\nabla z\|_{L^2}^2 \;\lesssim \int_{t_0}^t\!\!\int |z|^2\,|\nabla v|\,dx\,ds.$$

By local Sobolev and interpolation, $\int |z|^2|\nabla v| \lesssim \|\nabla v\|_{L^2}\,\|z\|_{L^2}^{1/2}\,\|\nabla z\|_{L^2}^{3/2} \le \varepsilon\|\nabla z\|_{L^2}^2 + C_\varepsilon\|\nabla v\|_{L^2}^4\,\|z\|_{L^2}^2$. Absorbing the ε-term and applying Grönwall (noting $\nabla v \in L^2_{\mathrm{loc}}(I; L^2)$ from mild theory on $(t_0, t_0+T]$), we conclude $z \equiv 0$ on I. □

6.4 Identification with the ancient profile and uniform iteration

Theorem 6.4 (Weak–strong identification and uniform iteration)**.** *Let U be the ancient limit from §4 and let $u^{(n)}$ be the dynamically centered rescalings with data $f_n := u^{(n)}(-1)$. Set $f := U(-1)$ and assume $\sup_n \|f_n\|_{L^3} \le M$. Then:*

1. ***First window.*** *$f_n \to f$ in L^3 (by §4, time continuity and uniform tails). By Prop. 6.2, there exist $T_0 = T_0(M) > 0$ and mild solutions v_n, V on $[-1, -1+T_0]$ with $\sup_t \|v_n(t) - V(t)\|_{L^3} \to 0$ and $\sup_n \|v_n\|_{L^\infty_t L^3_x} \lesssim M$. Lemma 6.3 gives $u^{(n)} \equiv v_n$ and $U \equiv V$ on $[-1, -1+T_0]$, hence $\sup_t \|u^{(n)}(t) - U(t)\|_{L^3} \to 0$ there.*

2. ***Uniform iteration.*** *Since $\|v_n(t_1)\|_{L^3} \le CM$ at $t_1 = -1 + T_0$, Prop. 6.2 restarts on $[t_1, t_1 + T_0]$ with the same T_0. Iterating K times yields existence and identification on $[-1, -1 + KT_0]$. Picking K minimal with $-1 + KT_0 \ge \tau > 0$ gives that, for large n, both $u^{(n)}$ and U exist (mild) and coincide on $[-1, \tau]$, with uniform convergence on each subwindow.*

6.5 ESS on the ancient profile and contradiction

Theorem 6.5 (ESS for U; contradiction)**.** *Assume u has a singular point at $(x_*, 0)$. Let $u^{(n)}$ be the centered rescalings and U the ancient limit with $u^{(n)} \to U$ strongly in L^3_{loc}. Then:*

1. *(*No-vanishing*) By §4 (contrapositive of one-scale ε–regularity), there exists $\varepsilon_* > 0$ such that*
$$\Phi_U(2r) \;\geq\; \tfrac{1}{2}\,\varepsilon_* \qquad \forall\, r > 0.$$

2. *(*Critical global bound*) By §4.2 we have $\sup_{t<0} \|U(t)\|_{L^3(\mathbb{R}^3)} \leq M_* < \infty$ and $U \in C((-\infty, 0]; L^3)$.*

3. *(*ESS at the boundary*) Applying Escauriaza–Seregin–Šverák (E5) to U on $(-T, 0]$ (any $T > 0$) gives smoothness up to $t = 0$ and forward extension.*

4. *(*Smallness at tiny scales*) Smoothness near $(0, 0)$ implies: for some $r_0 > 0$, $\Phi_U(2r_0) \leq \frac{1}{4}\,\varepsilon_*$ (take r_0 small so that $r_0^{-2} \iint_{Q_{2r_0}} |U|^3$ is tiny).*

Items (1) and (4) contradict. Hence $(x_*, 0)$ cannot be singular. Undoing time shift shows no blow-up at $t = 0$.

Non-circularity. The ESS step is used *only after* the independent global L^3 bound for the ancient profile U is established from the local block + dynamic compactness. No part of §4 uses ESS to obtain that bound, and all pressure effects are accounted for through normalized gauges and explicit corona budgets.

Summary of Section 5. *(i)* Prop. 6.1 furnishes the time-fractional gain with $\alpha = \delta/(3 + \delta)$. *(ii)* Prop. 6.2 yields a uniform-in-data existence time and Lipschitz L^3 stability. *(iii)* Thm. 6.4 synchronizes rescalings with the ancient profile on a first window and iterates to a common positive time. *(iv)* Thm. 6.5 then applies ESS to U to force smoothness/extension at $t = 0$, contradicting the inherited non-vanishing of Φ_U at all scales. The contradiction rules out finite-time singularities.

7 Global regularity

Geometry and averages. Fix $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and $r > 0$. Set $B_r := B_r(x_0)$, $I_r := (t_0 - r^2, t_0]$, $Q_r := B_r \times I_r$, and $\fint_E f := |E|^{-1} \int_E f$ for any measurable E. For the pressure, write the normalized gauge on Q_r:
$$\widetilde{p}(\cdot, t) := p(\cdot, t) - \fint_{B_r} p(\cdot, t)\, dx.$$

Scale–invariant functionals. All quantities below are dimensionless under the Navier–Stokes scaling $u_\lambda(x, t) = \lambda\, u(x_0 + \lambda x,\, t_0 + \lambda^2 t)$, $p_\lambda = \lambda^2 p$:
$$\Phi(r) := \fint_{Q_r} |u|^3, \qquad \Psi(r) := \fint_{Q_r} |\widetilde{p}|^{3/2},$$
$$E(r) := \sup_{t \in I_r} \fint_{B_r} |u(\cdot, t)|^2, \qquad F(r) := r \fint_{Q_r} |\nabla u|^2,$$
$$\mathrm{Ent}(r) := \fint_{Q_r} (\rho - \rho_{Q_r})^2, \qquad \rho := \log(1 + |u|/k),\ k > 0, \qquad \mathcal{M}(r) := F(r) + \lambda\, \mathrm{Ent}(r).$$

For $\theta \in (0, 1)$ we also use the outer tail $\mathrm{Tail}_\theta(r) := r^{-2} \iint_{Q_r \setminus Q_{\theta r}} \big(|u|^3 + |\widetilde{p}|^{3/2}\big)$, which is $\downarrow 0$ as $\theta \uparrow 1$.

Theorem 7.1 (Global regularity)**.** *Let $u_0 \in \mathcal{S}(\mathbb{R}^3)$ be divergence-free and let (u, p) be a suitable (Leray–Hopf + LEI) solution to the 3D incompressible NSE on $\mathbb{R}^3 \times [0, \infty)$ with $u(\cdot, 0) = u_0$ and no forcing. Then u is smooth on $\mathbb{R}^3 \times (0, \infty)$.*

Proof. **Step 0 (Contradiction setup and normalization).** Assume there exists $T > 0$ and $x_* \in \mathbb{R}^3$ such that (x_*, T) is singular. By space–time translation we may assume $(x_*, T) = (0, 0)$ and argue on $(-T_0, 0]$ for some $T_0 > 0$ to be fixed.

Step 1 (Local one–scale ε–regularity engine). Section 2 establishes the local block: inter–scale Caccioppoli (Lemma 3.1), log–Caccioppoli + hole filling (Lemma 3.2), reverse Hölder/Gehring (Lemma 3.4), the local $L^{3+\delta}$ gain (Lemma 3.5), and the critical Moser iteration (Proposition 3.6). As a consequence, the one–scale ε–regularity holds: there exists a universal $\varepsilon_* > 0$ such that

$$\Phi(2r) \le \varepsilon_* \quad \Longrightarrow \quad u \text{ is smooth in } Q_{r/2}. \tag{7.1}$$

(See Theorem 3.8.)

Step 2 (Good–λ descent for the defect $\mathcal{M}$ and bridge to Φ). In Section 3 we introduced the defect $\mathcal{M} = F + \lambda$ Ent and proved its discrete contraction across scales: for each fixed $\theta \in (0, 1)$ there exist $\kappa(\theta) \in (0, 1)$ and $C_\theta < \infty$ such that

$$\mathcal{M}(\theta r) \ \le \ \kappa(\theta)\, \mathcal{M}(r) \ + \ C_\theta\, [\Phi^{1/3} E + \Phi + \Psi](r). \tag{7.2}$$

We also proved the bridge that upgrades $\mathcal{M}$ back to Φ at a smaller scale:

$$\Phi(\theta r) \ \lesssim_\theta \ \Phi(r) + \Psi(r) \ + \ \mathcal{M}(r)^{3/2} \ + \ (\Phi(r) + \Psi(r))^{3/2}. \tag{7.3}$$

The one–step form (7.2) with cutoffs supported in the corona and a packing argument imply an explicit corona budget and a finite–step descent: for radii $r_j = \theta^j r_0$,

$$\mathcal{M}(r_J) \ \lesssim \ (1 - c)^J\, \mathcal{M}(r_0) \ + \ C\, \big[\Phi(2r_0) + \Psi(2r_0)\big] \tag{7.4}$$

for some universal $c \in (0, 1)$. When the exterior budget $\Phi(2r_0) + \Psi(2r_0)$ is small, (7.3) then forces $\Phi(2r_{J+1}) \le \varepsilon_*$, and (7.1) yields local smoothness at that smaller scale.

Step 3 (Dynamic compactness, ancient profile, and non–vanishing). We do *not* assume small exterior budget globally. Instead, we follow the dynamics with a canonical gauge as in Section 4: for each $t < 0$ pick a center $x(t)$ and a median radius $\lambda(t) > 0$, and define the rescaling

$$u^{(n)}(x, t) \ := \ \lambda(t_n)\, u\big(x(t_n) + \lambda(t_n) x, \ t_n + \lambda(t_n)^2 t\big), \qquad t_n \uparrow 0.$$

The APMS (all–parabolic–multi–scales) Carleson bound for F (from the good–λ descent with anchored pressure) and a terminal slicing lemma (endpoint LEI test with time cutoff) give uniform *pointwise–in–time* control of far–field L^3 tails in the gauge. A space translation modulus (via the local L^q with $q > 3$ from the Moser step and H^1–type control) and a time modulus (from $\partial_t u = \Delta u - \mathbb{P}\nabla \cdot (u \otimes u)$ in a negative Sobolev space, interpolated with the L^q gain) yield compactness: up to a subsequence,

$$u^{(n)} \ \to \ U \quad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0]). \tag{7.5}$$

The limit U is a suitable ancient solution. Moreover, the tail control passes to the limit, and an interpolation on balls shows

$$\sup_{t \in (-T, 0]} \|U(t)\|_{L^3(\mathbb{R}^3)} \ < \ \infty \tag{7.6}$$

for some $T > 0$ independent of n. Finally, the contrapositive of the one–scale ε–criterion (7.1) and the invariance of Φ under scaling show that U is *non–vanishing* at every scale:

$$\Phi_U(2r) \ \ge \ \tfrac{1}{2}\, \varepsilon_* \qquad \forall\, r > 0. \tag{7.7}$$

Step 4 (Kato L^3 window and weak–strong identification). Let $f_n := u^{(n)}(-1)$ and $f := U(-1)$. The compactness framework gives $f_n \to f$ in $L^3(\mathbb{R}^3)$. Kato's critical L^3 theory

provides a uniform existence time $T_0 = T_0(\sup_n \|f_n\|_{L^3}) > 0$ and unique mild solutions v_n, V on $[-1, -1+T_0]$, with Lipschitz dependence on data. A local weak–strong argument with cutoffs and the uniform L^3 tails identifies suitable and mild solutions on this window: $u^{(n)} \equiv v_n$ and $U \equiv V$ on $[-1, -1+T_0]$. Iterating the window (since $\|v_n(-1+T_0)\|_{L^3}$ is uniformly controlled) yields, for some $\tau > 0$, that both $u^{(n)}$ and U are mild and coincide on $[-1, \tau]$ for n large. This preserves (7.6) and continuity up to $t = 0$ for U in L^3.

Step 5 (ESS rigidity at the boundary and contradiction). By (7.6), the Escauriaza–Seregin–Šverák boundary regularity criterion applies to U on $(-T, 0]$: U is smooth up to $t = 0$ and extends for $t > 0$. Smoothness near $(0, 0)$ implies small *critical mass* at small scales: there exists $r_0 > 0$ such that

$$\Phi_U(2r_0) \;\le\; C\,\|U\|^3_{L^\infty(Q_{2r_0})}\, r_0^3 \;\le\; \tfrac{1}{4}\,\varepsilon_*$$

for r_0 sufficiently small. This contradicts the non–vanishing lower bound (7.7), which states $\Phi_U(2r) \ge \frac{1}{2}\,\varepsilon_*$ for all $r > 0$. Therefore the assumption of a singular point at $(0, 0)$ is false.

Step 6 (Global smoothness). Since $T > 0$ was arbitrary, there are no finite–time singularities. With $u_0 \in \mathcal{S} \subset L^3$, Kato's theory gives a mild solution that is smooth for small positive times; the absence of singular times implies u extends smoothly for all $t > 0$. □

Non–circularity and bookkeeping. The global L^3 bound (7.6) for the ancient U is proved *without* ESS; ESS is applied only afterwards to U. The nonlocal pressure is handled by the normalized gauge and tails, and every cutoff derivative is included in the corona budget. All identities are adimensional and invariant under parabolic scaling.

8 Conclusions

The argument demonstrates that, for smooth rapidly decaying divergence–free data on $\mathbb{R}^3$ and no forcing, incompressible NSE yields a global smooth flow; via weak–strong identification, uniqueness holds within the suitable class. Beyond the theorem, the proof's architecture is reusable. Three principles are portable across critical PDEs:

- **Scale–invariant bookkeeping.** Measure what the equation measures. Instead of raw norms, work with dimensionless packages (e.g., critical mass, dissipation) augmented by a convex "tax" that penalizes concentration where it hurts (here, a log–entropy).
- **Defect–plus–descent.** Build a defect that contracts across scales under a good–λ scheme. Each step forces a dichotomy—either the ε–regularity trigger already fires, or the defect drops by a fixed factor—so a good scale appears in finitely many steps.
- **Dynamic compactness plus boundary rigidity.** Use a canonical gauge (centers and scales tied to the critical distribution) to prevent loss of mass at infinity, extract an ancient profile, and then close with the appropriate rigidity at the boundary of time.

These ideas suggest a template in other critical settings: in dispersive models, a Strichartz–plus– Morawetz defect with a space–time good–λ and a scattering/virial rigidity; in energy–critical wave or geometric flows, localized virials/monotonicity playing the rigidity role while a corona budget accounts for cutoff losses. The governing rule is simple: never borrow regularity from the future—keep estimates local, count every boundary term, and invoke rigidity only when the hypothesis is already secured by independent means.

If taken at face value, a global regularity theorem for incompressible NSE says that, in the continuum limit we use to model fluids, the future is *single–valued*: from smooth, rapidly decaying initial data there is one, globally defined, smooth trajectory for all positive time, and no hidden singular fork where the evolution could branch. In this precise mathematical sense, the

apparent indeterminacy of turbulence is epistemic (resolution, data, numerics), not ontological. At the same time, the path to this conclusion—fifteen years of assembling local truths, counting every boundary term, and refusing to borrow regularity from the future—carries a philosophical echo: the present contains enough information, once expressed in the right scale–invariant currencies, to forbid rupture. This does not legislate on human freedom or metaphysical will; it calibrates what "determinism" means in one of our most universal physical models. The equations admit complexity without catastrophe; they allow unpredictability in practice without ambiguity in principle. That, perhaps, is the deeper lesson: when the grammar of a problem matches its symmetries, the worldline does not bifurcate—the mystery of the cascade survives, but its logic remains intact.

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